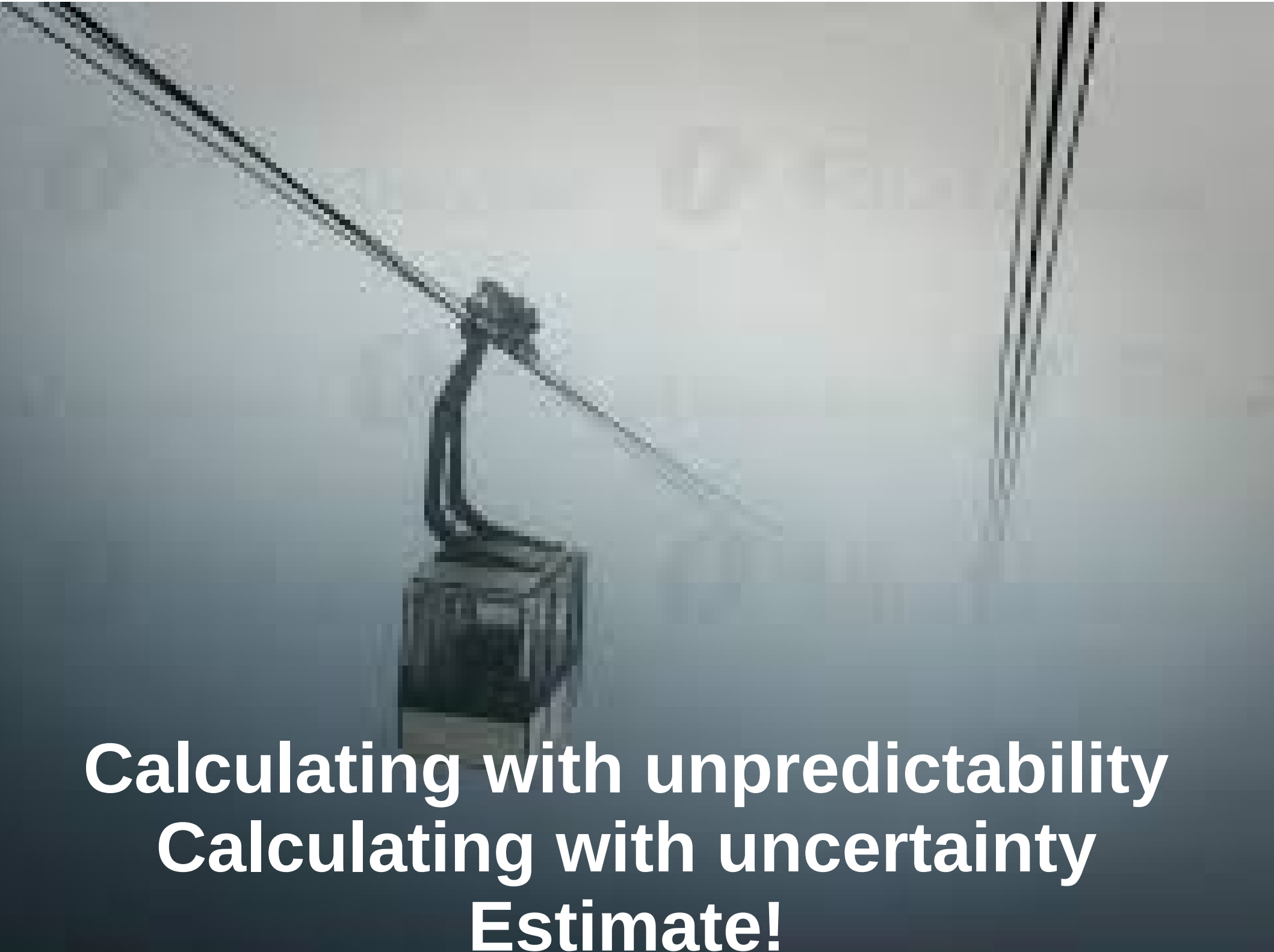


Calculating with unpredictability
Calculating with uncertainty
Estimate!

How many sensors do we have to combine to get the current state?

Calculating with uncertainty (certainty und uncertainty)

1 + 3 = 4,0000000000000000000000000000000000000000000000000000000000000000...0

This exists only in mathematics…

In physical systems, we always have uncertainty.
(We know the physical universe only as an approximation.)

1 + aprox. 3 = ?

ca. 1 + aprox. 3 = ?

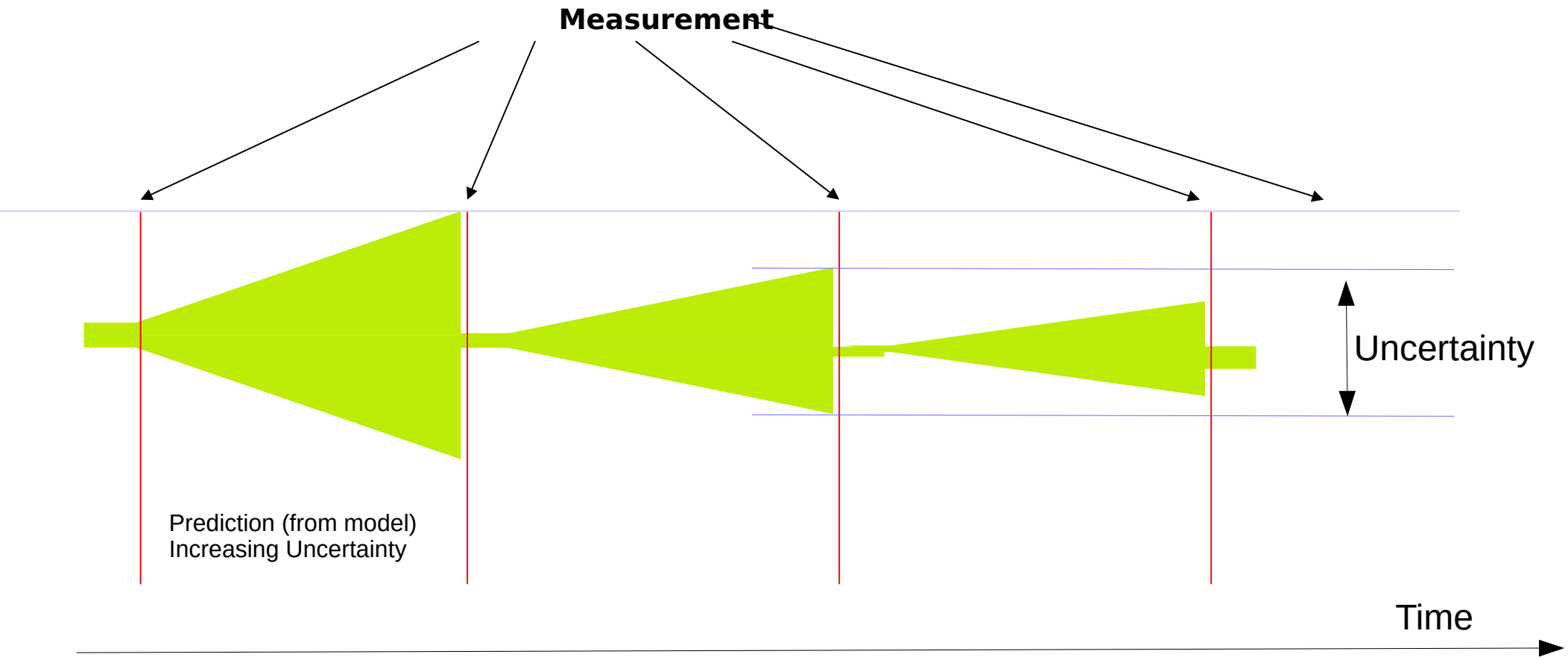
$$\frac{1}{aprox.0} = ¿$$

Calculating with uncertainty

Uncertainty = 1 - Certainty

	Value	Certainty (0..1)
1+3		
1+ aprox. 3		
aprox. 1 + aprox. 3		
1 / aprox. 0		

Model/Prediction + Measurements



Model/Prediction + Measurements

Remember:

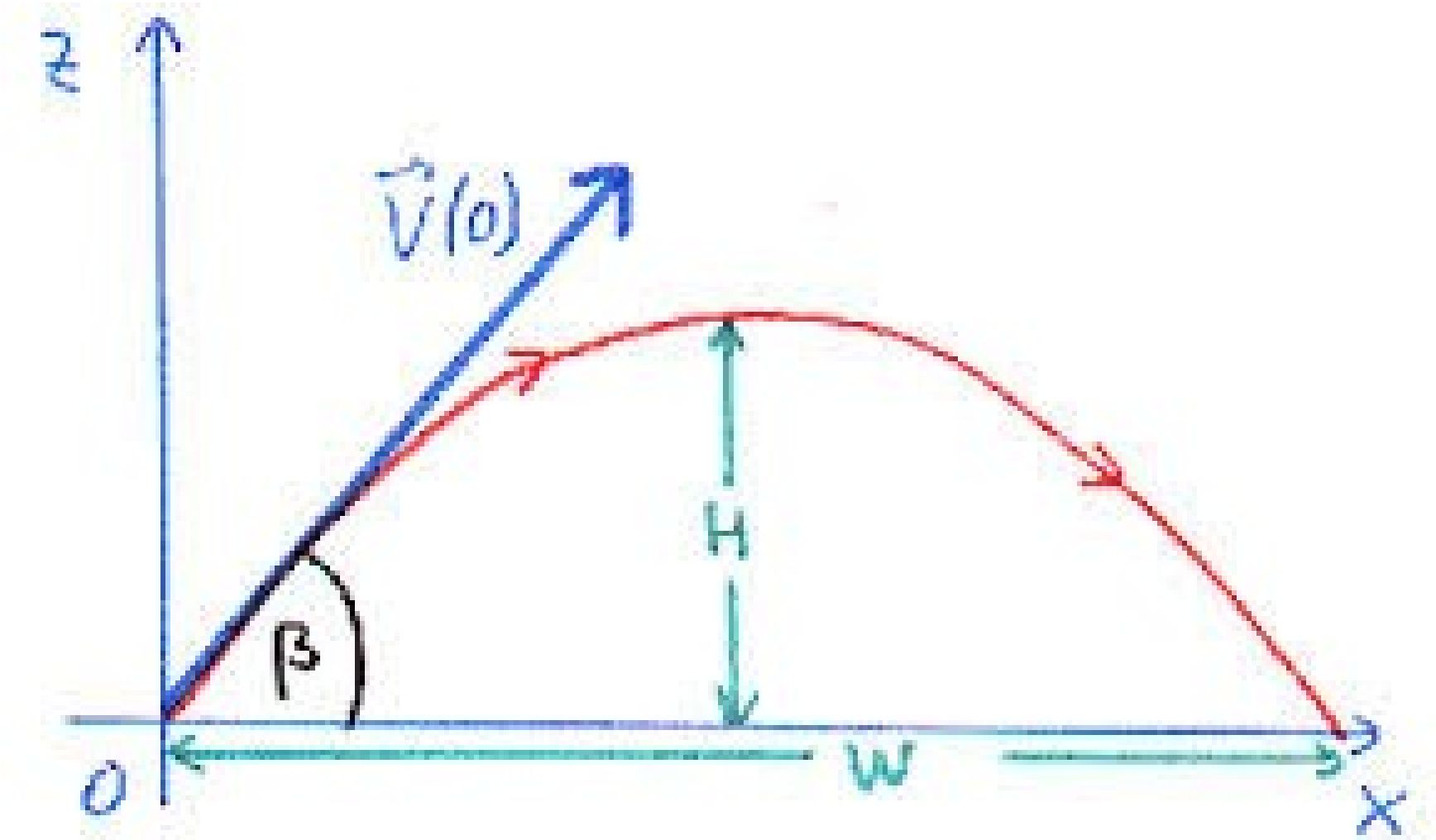
Data from sensors need to be filtered (reduce noise), and show the past
From the model we get (fast) the present and future situation

→ from the model the high frequency & present (& future) situation.
→ from the sensors the tendency and corrections

But for simple, slow systems

I can take the measurements as treu value

Why Sensors?
I can calculate!



Why Models and Calculations? I can measure!

Speaking of... Am I really measuring the temperature? Or what am I looking at? A model!

Both: Model and Measurement have Uncertainty

$$\frac{\partial T}{\partial t} + \operatorname{div}(\vec{v}T) = \frac{k_l}{\rho_l c_l}\Delta T$$

$$\rho_l\left(\frac{\partial \vec{v}}{\partial t} + \operatorname{div}(\vec{v}\otimes\vec{v})\right) = -\operatorname{grad} p + \mu\Delta\vec{v} + \rho_l\left(1-\beta(T-T_m)\right)\vec{g}$$

$$\operatorname{div}\vec{v} = 0$$

$$\frac{\partial T}{\partial t} = \frac{k_s}{\rho_s c_s}\Delta T$$

$$\left[k\frac{\partial T}{\partial n}\right]^l_s + \rho_l q(T-T_m)(\vec{V}_\Gamma - \vec{v})\cdot\vec{n} - \rho_s q(T-T_m)\vec{V}_\Gamma\cdot\vec{n} = -\rho_s L V_\Gamma,$$

$$\frac{T-T_m}{T_m} = -\frac{q\rho_s T_m V_\Gamma}{L} - \frac{\gamma\kappa_l}{L\rho_l} + \left[\frac{k^{-1}}{\rho}\frac{B-p^*}{L}\right]_s$$

$$\vec{v}(\vec{x}) = \left(1-\frac{\rho_l}{\rho_s}\right)\vec{V}_\Gamma, \qquad \vec{x}\in\Gamma_1$$

$$T_l = T_s,$$

$$k\frac{\partial T}{\partial n}(\vec{x}) = -h(T(\vec{x})-T^*), \qquad \vec{x}\in\Gamma_2$$

$$-p\vec{n} + \tau\vec{n} = -\gamma\kappa_l\vec{n} - p^*\vec{n}$$

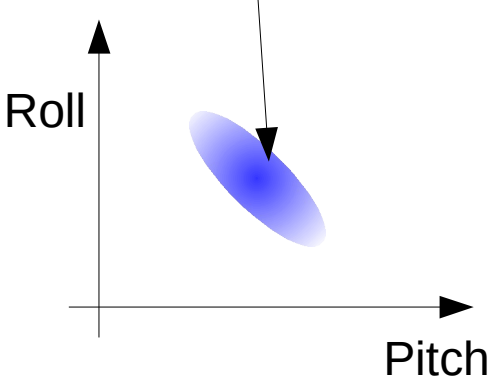
$$\vec{v}(\vec{x}) = 0, \qquad \vec{x}\in\Gamma_3$$

$$T(\vec{x}) = T_3(\vec{x}), \qquad \vec{x}\in\Gamma_3$$

$$k\frac{\partial T}{\partial n}(\vec{x}) = q_3(\vec{x}), \qquad \vec{x}\in\Gamma_3$$

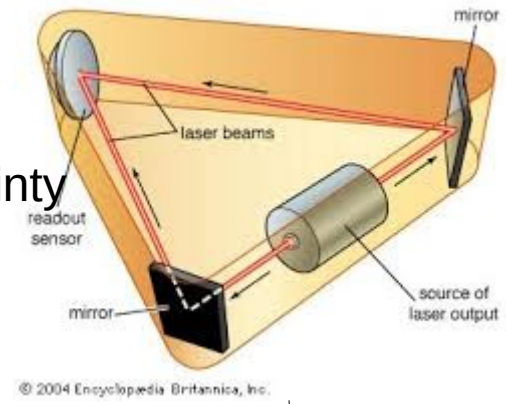
Where does uncertainty come from?

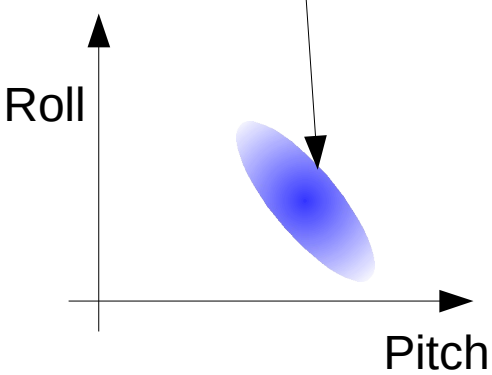
- Model inaccuracies
  - impossible to grasp physics completely
- Approximations
  - (Linearization)
- Unpredictables
- Initial state
- Time steps (discretization)
- Calculation accuracy
- ...



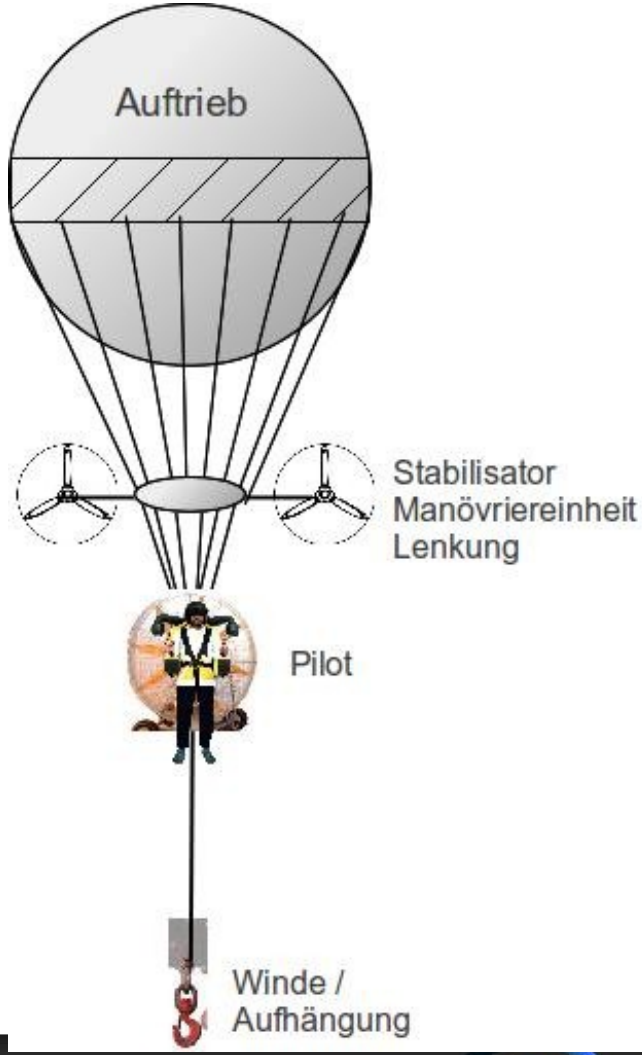
Where does uncertainty come from?

- Offset
- Noise
  - Electrical
  - Mechanical
- Drift (Random walk)
- Approximations
- Linearity





Is my model reliable and complete?



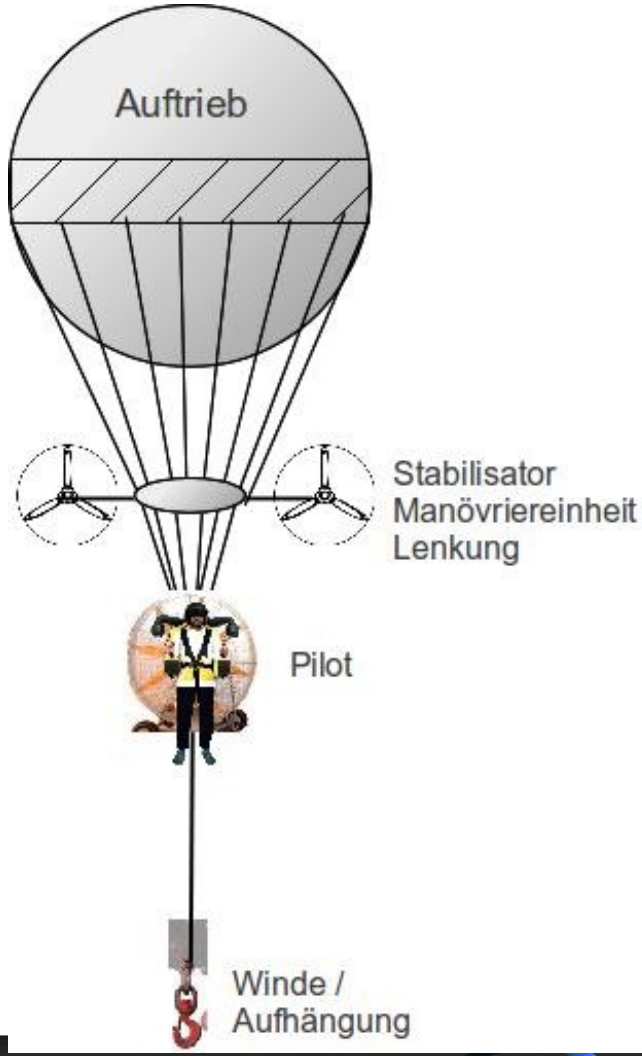
Is my model reliable and complete?

Does it take everything into account?

- Friction?
- Wind?
- Rain?
- Thermal expansion?
  - Temperature light/shadow? (clouds)
  - Air pressure & air pressure changes
- Thermals?
- Oscillations?
- Pilot's weight? His movements?

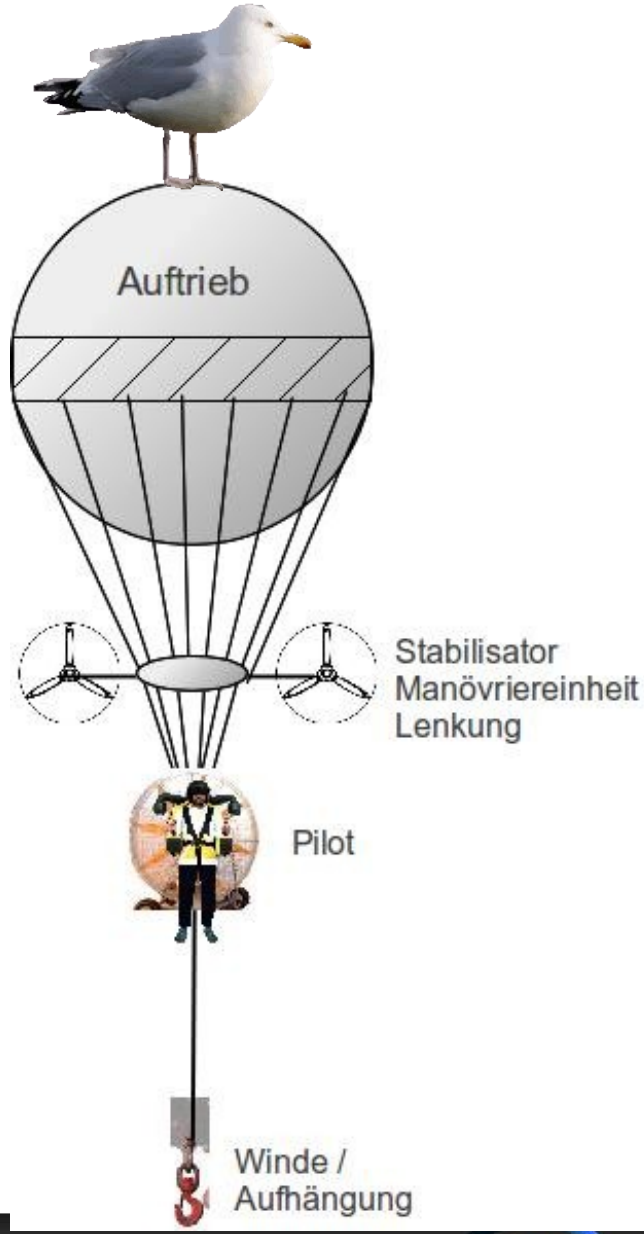
....

Certainly not!
There are many unpredictable parameters
and many unknown ones!



Is my model reliable and complete?

Does it take everything into account?



"Uncertainty" in measuring

Please „estimate“

honestly…
What do we know and how certain are we?

My pos = 100 cm
(sure?)

My pos = 100 cm

well, okay, +/- 10 cm

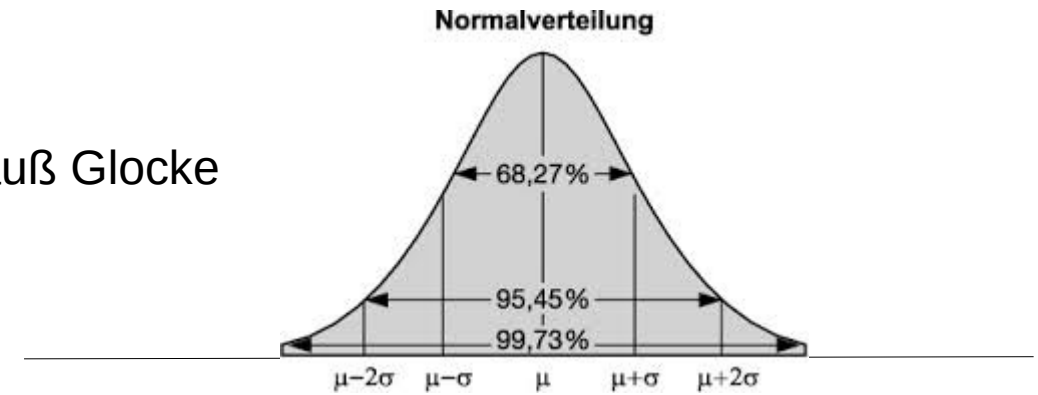
so 90 .. 110 cm.

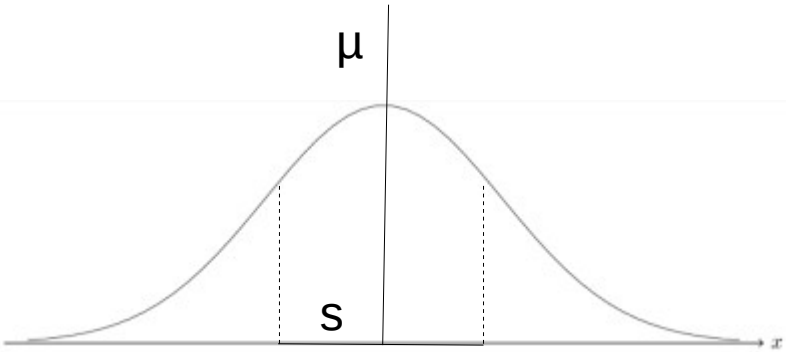
(111 cm is impossible?)

Gaussian distribution

Probability of my position (please integrate, from... to. This gives me certainty for this area) :

Gauß Glocke





Standard deviation = 1-Sigma

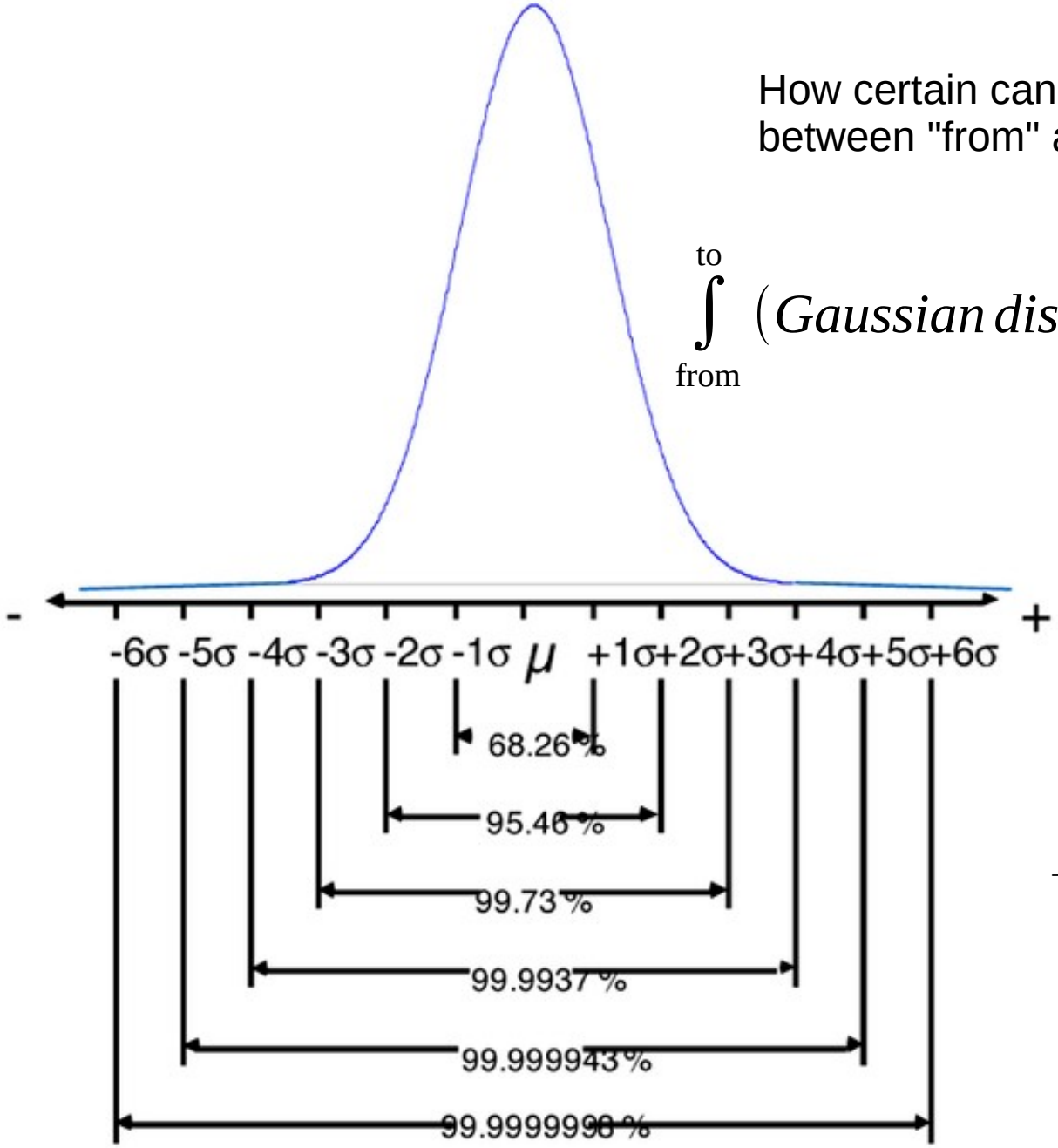
My pos = 100 cm
well, okay, then Gaussian distribution
x = μ = 100 cm
Standard deviation (sigma) = 5 cm
z.B.

+/- 5 cm bei 1-Sigma (68% Certainty)
+/- 10 cm bei 2-Sigma (95% Certainty)
+/- 15 cm bei 3-Sigma (99,7% Certainty)

Certainty at 100 cm +/- 0 ?

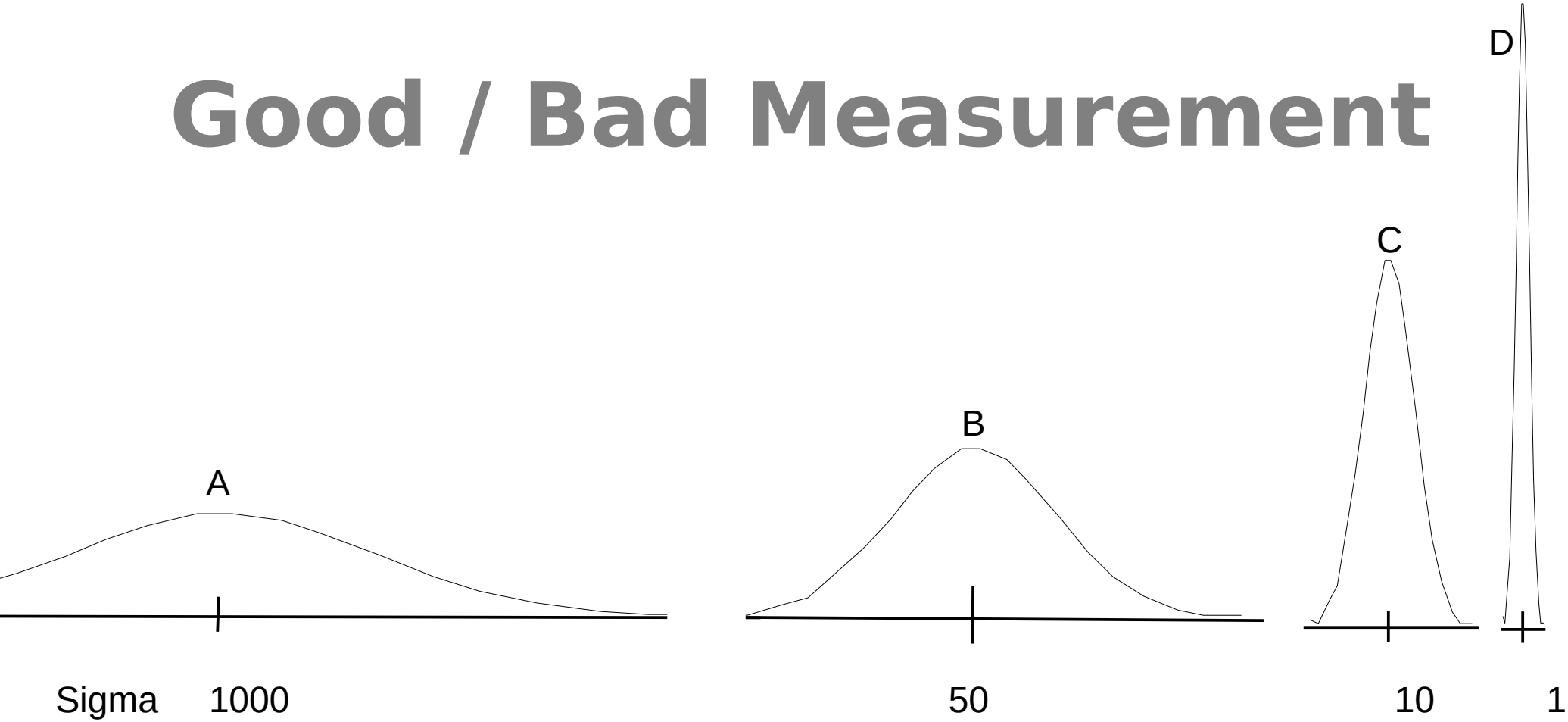
How certain can I be that the real value is between "from" and "to"?

$$\int_{\text{from}}^{\text{to}} (Gaussian\ distribution) = My\ cerntainty$$



$$\int_{-\infty}^{\infty} (Gaussian\ distribution) =$$

Good / Bad Measurement



When am I more certain of my measurement? Higher certainty?
How wide is 1-sigma? e.g. +/- N cm, but these are only 68% of the cases.

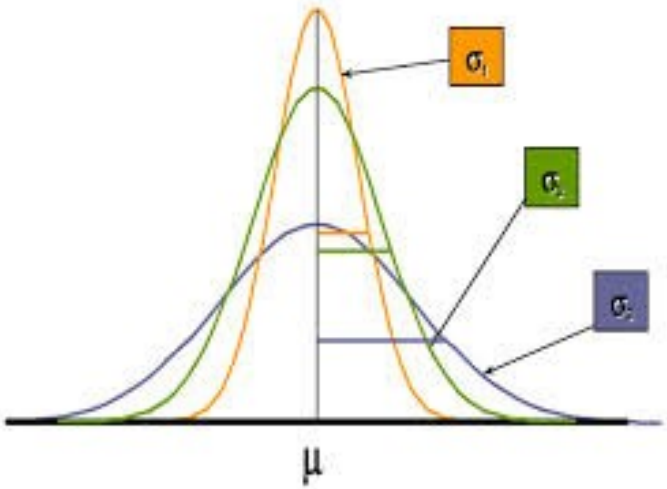
Remember

$$average = \mu = \bar{x} = \frac{1}{n} * \sum x_i$$

$$Varainz = v = \frac{1}{n} * \sum (x_i - \bar{x})^2$$

$$Standar\ deviation = s = \sqrt{v}$$

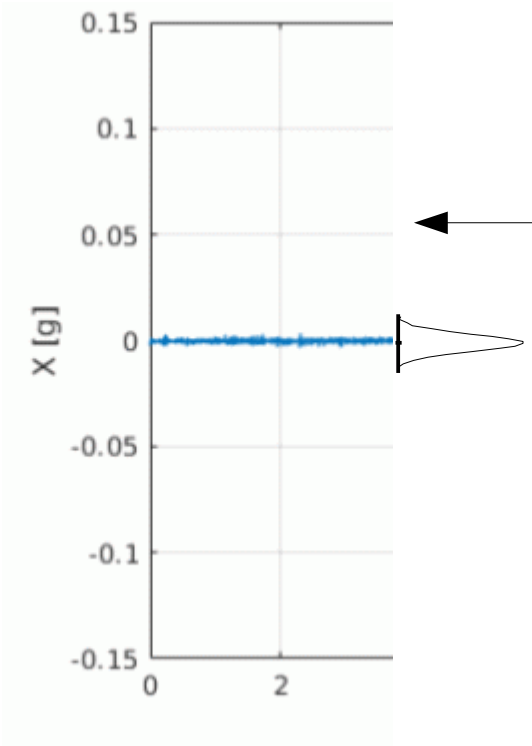
$$Contour = f(x) = \frac{e^{-(x-\bar{x})^2/2*v^2}}{\sqrt{2*\pi*v}}$$



Probability of von „exactly a“ = 0
at maximum uncertainty (I don‘t know anything),
s = infinite,
f(x) = a horizontal line very near 0
Absolutely certain of my statement (does not exist!):
s = 0
f(μ) = infinitely high, f(x != μ) = 0

$$\text{Probability that I am between a and b} = \int_a^b f(x)$$

Good / Bad Measurement



Acceleration sensor

Which sensor is better?

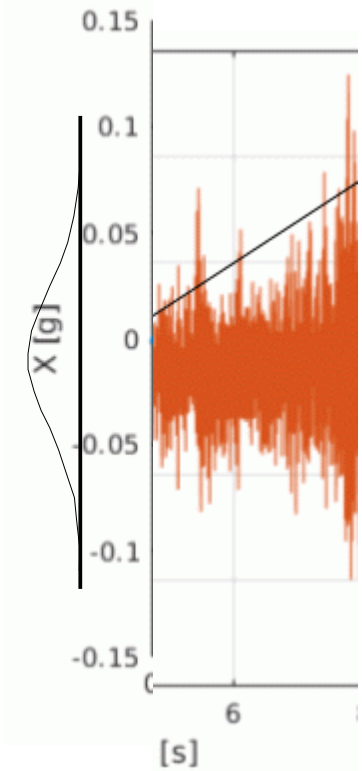
How does Gaussian distribution interpret the blue and the orange line?

μ = ?

s = ?

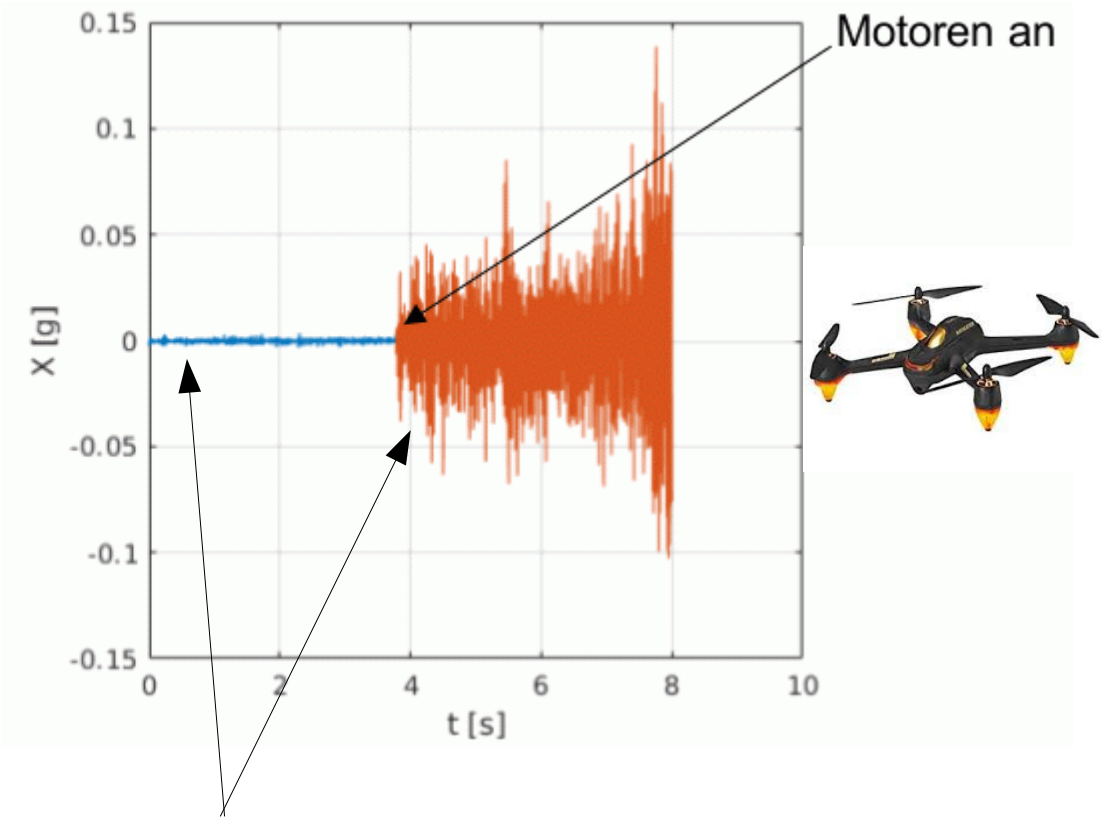
I want to know it with +/- 0,001 m/s²

Certainty (0..1) x = 0 +/- 0,001 = ?



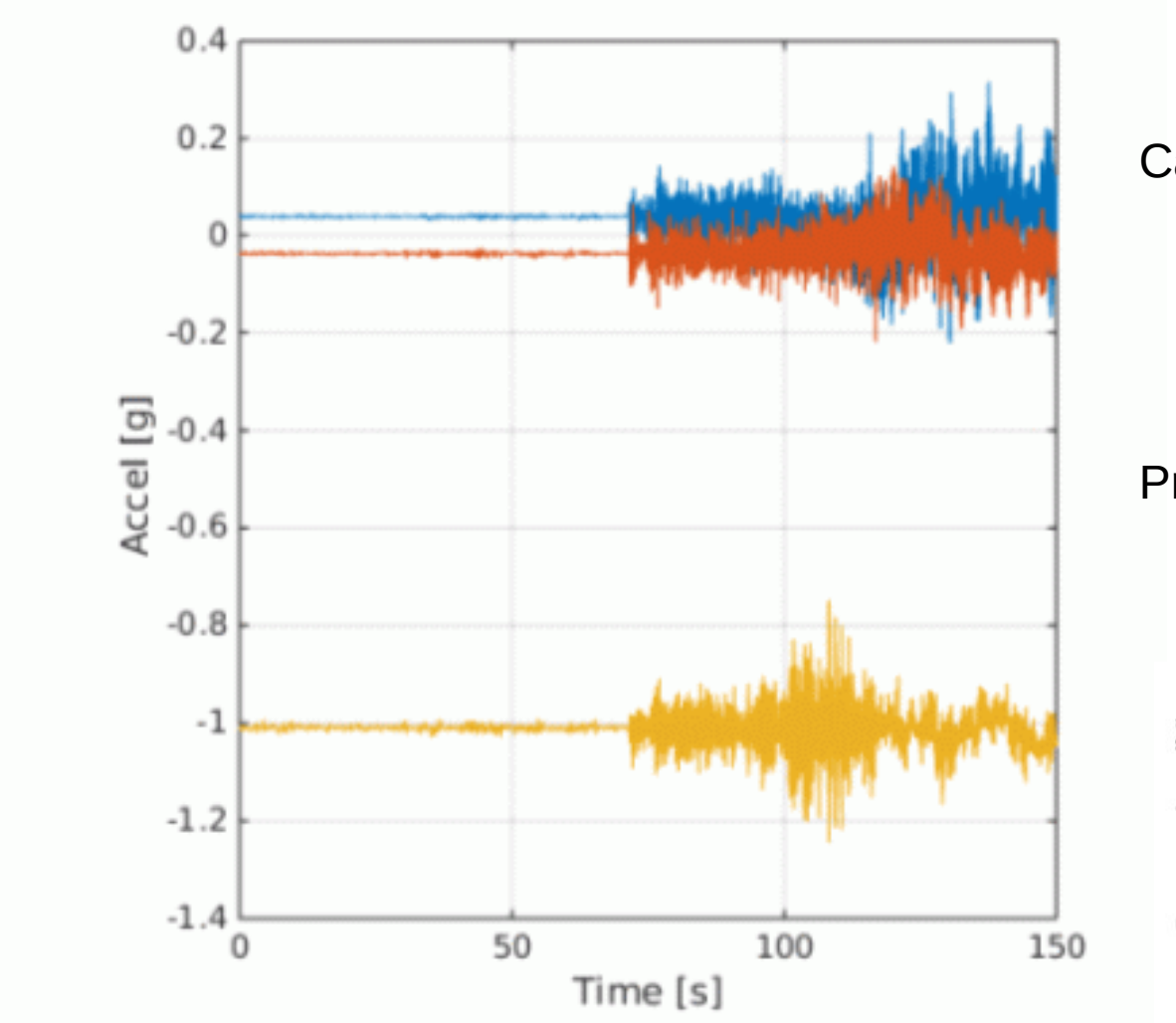
Good / Bad Measurement

The same sensor may produce good and bad measurements.



This noise - product
of sensor? Or wherefrom?

Offset, Drift, Noise

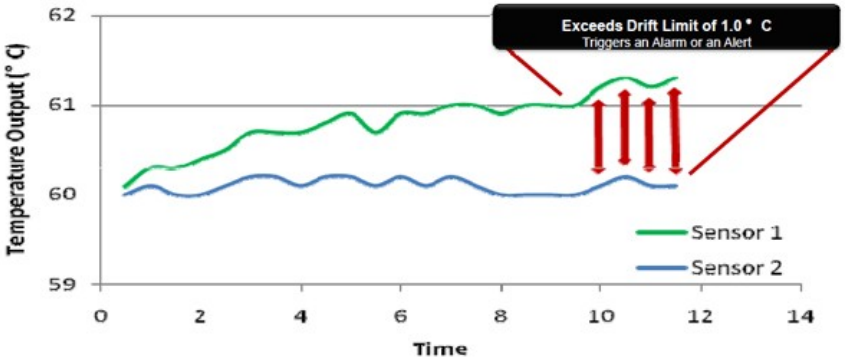


Calibration: (IMU calib. airplane: 10 minutes)

- Determine
  - Offset,
  - Noise distribution (changes!)
  - Drift

Problem: Drift „Random Walk“

If it weren't for "random" I would be able to model it!



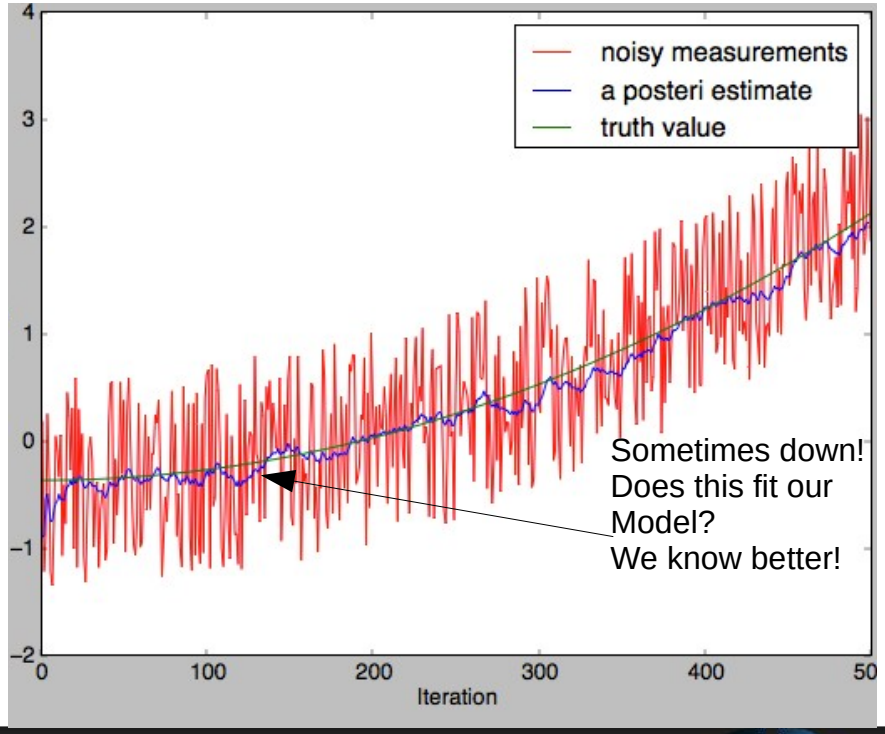
Temperature did not change the whole time

Is my Measurement reliable?

Launchers: Acceleration : 0 .. 5 g (typical < 3g)
Vibrations: 100g .. 1K g
What am I measuring then?

$$h = \iint a_v$$

Integration accumulates error!

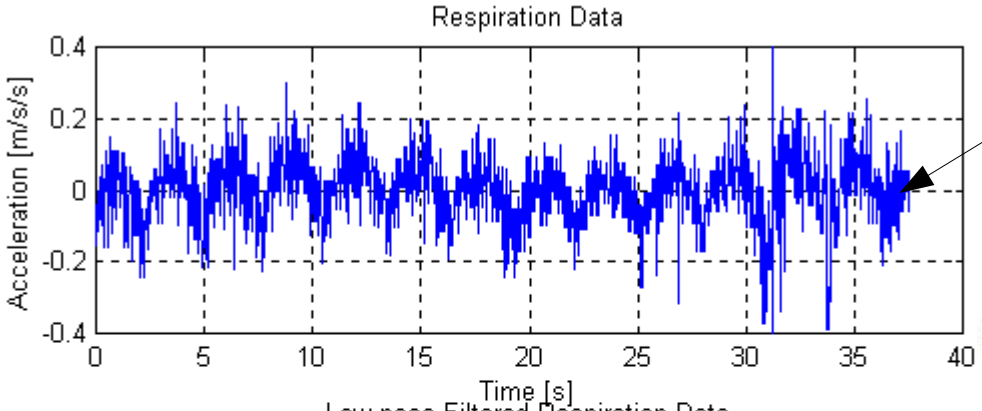


→ Filter, omit high frequencies
(disadvantage: I only notice tendencies)

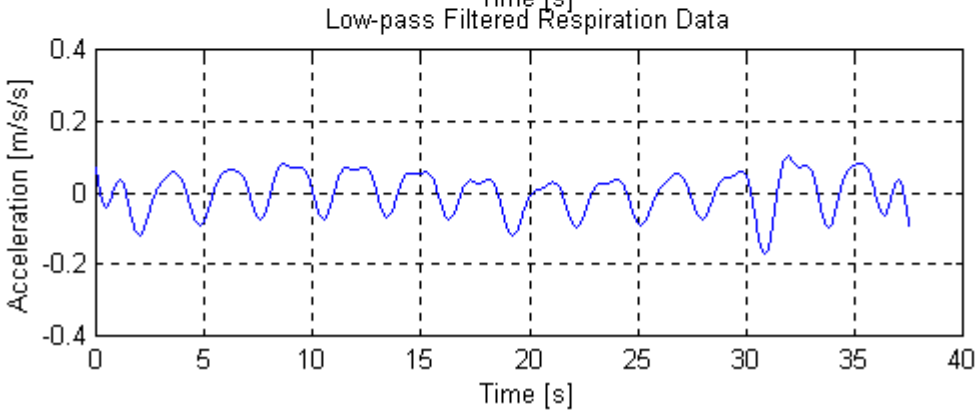
Filter: Ignore Measurements?

Filter: Ignore Measurements?

Acceleration, but vibration has to be omitted



These high frequencies were there!
They are real!
But I don‘t want to visualize them



Average of the last 100 measurements?
-> Low pass filter:
High frequencies disappear

Low pass filter: I am looking into the past!

Both: Model and Measurement have Uncertainty

$$\frac{\partial T}{\partial t} + \operatorname{div}(\vec{v}T) = \frac{k_l}{\rho_l c_l}\Delta T$$

$$\rho_l\left(\frac{\partial \vec{v}}{\partial t} + \operatorname{div}(\vec{v}\otimes\vec{v})\right) = -\operatorname{grad} p + \mu\Delta\vec{v} + \rho_l\left(1-\beta(T-T_m)\right)\vec{g}$$

$$\operatorname{div}\vec{v} = 0$$

$$\frac{\partial T}{\partial t} = \frac{k_s}{\rho_s c_s}\Delta T$$

$$\left[k\frac{\partial T}{\partial n}\right]^l_s + \rho_l a(T-T_m)(\vec{V}_\Gamma - \vec{v})\cdot\vec{n} - \rho_s a_s(T-T_m)\vec{V}_\Gamma\cdot\vec{n} = -\rho_s L V_\Gamma$$

$$\frac{T-T_m}{T_m} = -\frac{a\rho_s T_m V_\Gamma}{L} - \frac{\gamma_l \kappa_l}{L\rho_l} + \left[\frac{1}{\rho}\right]^{-1}_s \frac{p_l - p^*}{L}$$

$$\vec{v}(\vec{x}) = \left(1-\frac{\rho_l}{\rho_s}\right)\vec{V}_\Gamma, \qquad \vec{x}\in\Gamma_1$$

$$T_l = T_s$$

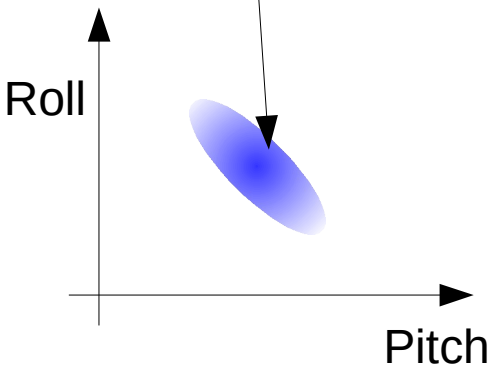
$$k\frac{\partial T}{\partial n}(\vec{x}) = -h(T(\vec{x})-T^*), \qquad \vec{x}\in\Gamma_2$$

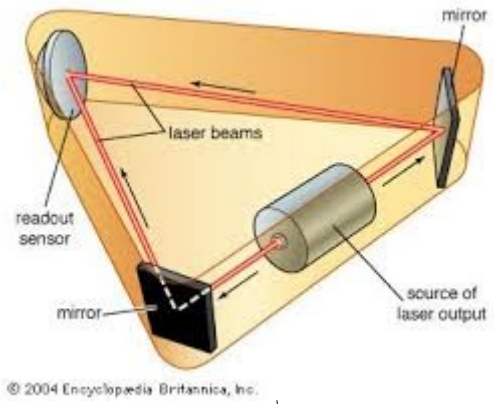
$$-p\vec{n} + \tau\vec{n} = -\gamma_l\kappa_l\vec{n} - p^*\vec{n}$$

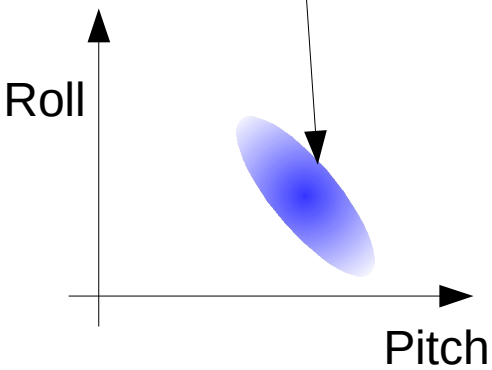
$$\vec{v}(\vec{x}) = 0, \qquad \vec{x}\in\Gamma_3$$

$$T(\vec{x}) = T_3(\vec{x}), \qquad \vec{x}\in\Gamma_3$$

$$k\frac{\partial T}{\partial n}(\vec{x}) = q_3(\vec{x}), \qquad \vec{x}\in\Gamma_3$$

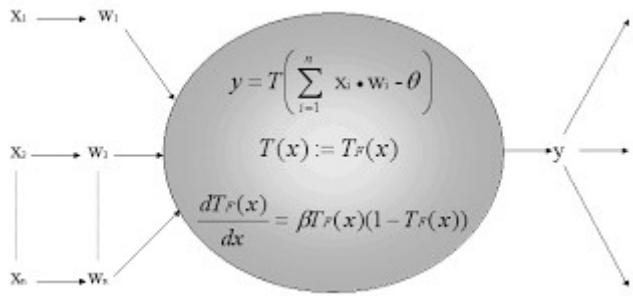


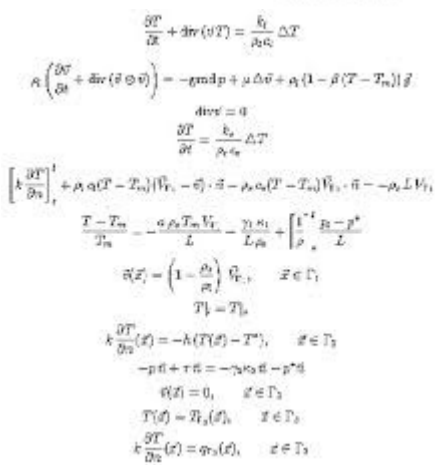


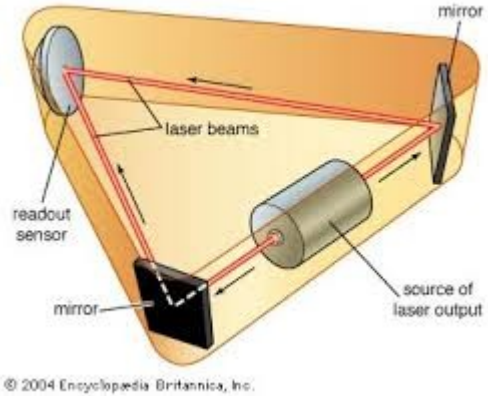


State estimation

Model (Simulate) + Sensors (Measure)





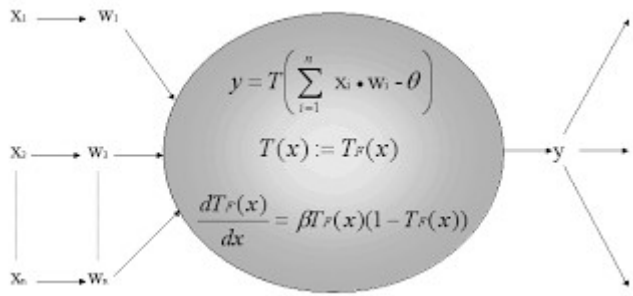


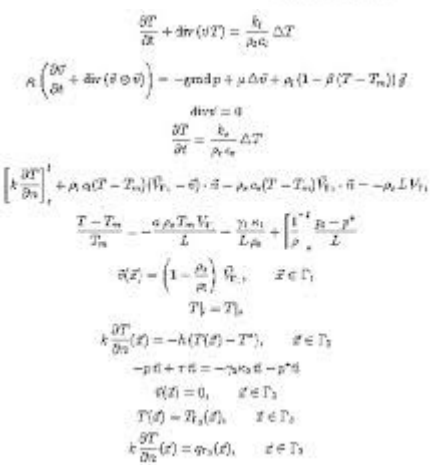
Filter

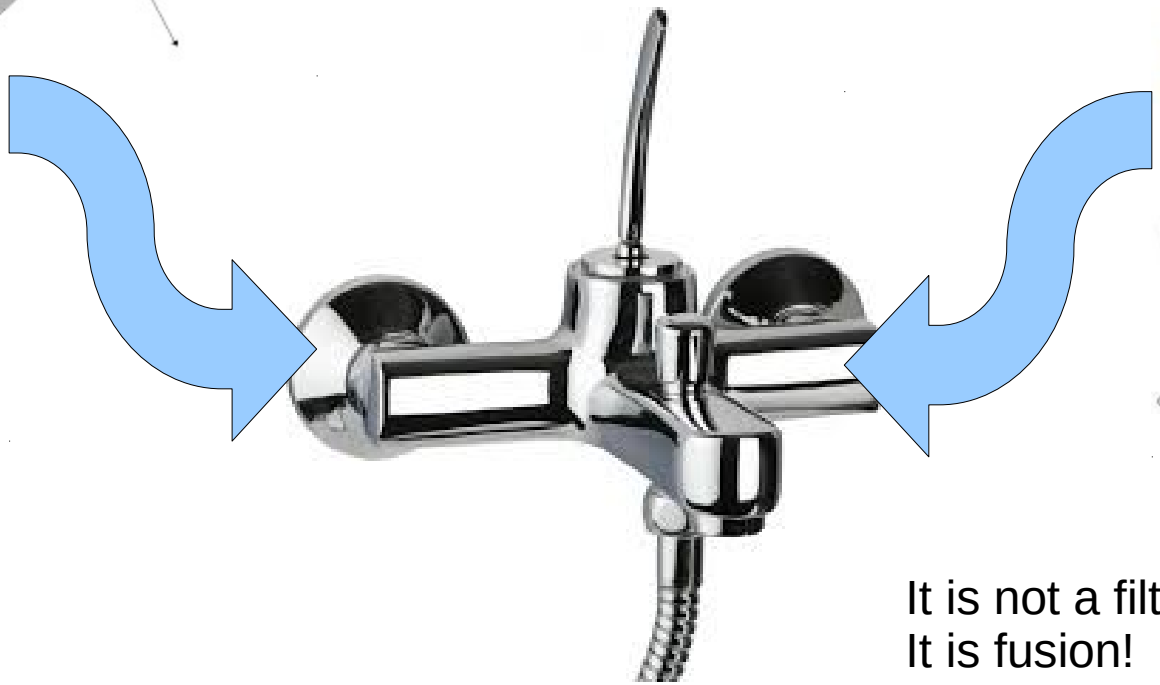
Filter (??)
Omit something, to be able to see the important things
e.g. omit vibrations, so I may see acceleration

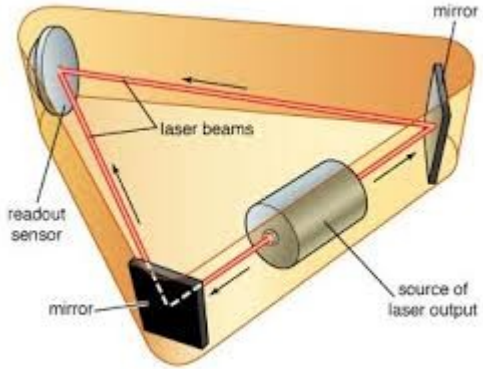
State estimation

Model (Simulate) + Sensors (Measure)









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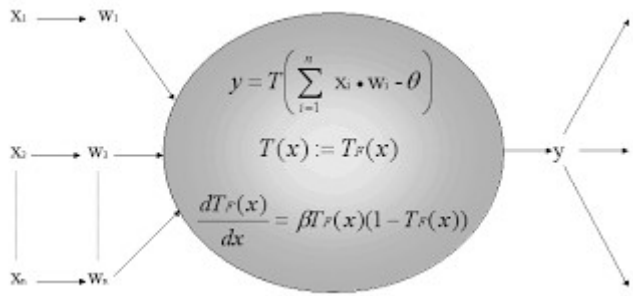
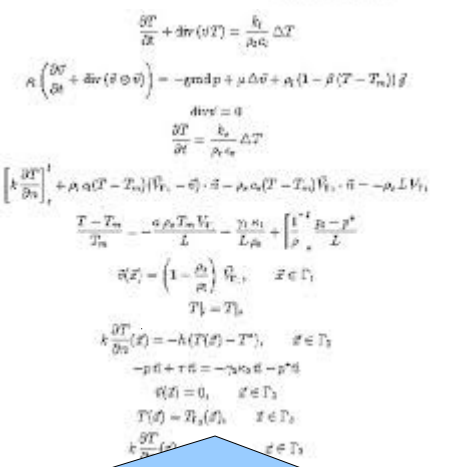
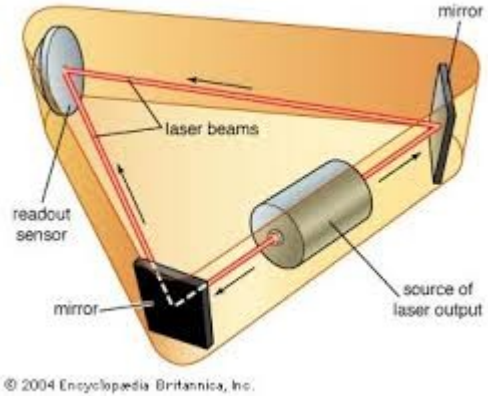
It is not a filter!
It is fusion!

State estimation

Model (Simulate) + Sensors (Measure)







„Best estimate“

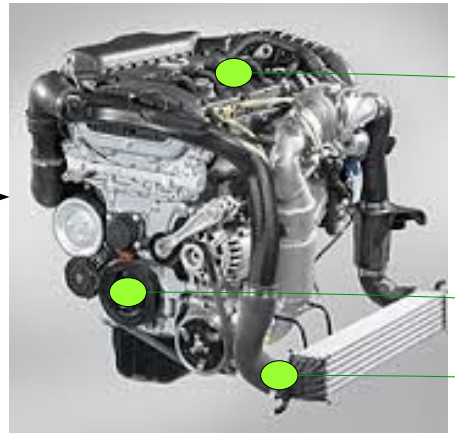
+ e. g. low pass filter
or frequency filter
etc.

Model + Measurements -> more!

```
Loop {
  Predict();
  correct();
}
```

= Sensor

Stimuli



control

Prediktor

Stimuli

$$\frac{\partial T}{\partial t} + \mathrm{div}(\vec{v}T) = \frac{k_f}{\rho_f c_f}\Delta T$$

$$\rho_f\left(\frac{\partial \vec{v}}{\partial t} + \mathrm{div}(\vec{v}\otimes\vec{v})\right) = -\rho_f \mathrm{grad}\, p + \mu\Delta\vec{v} + \rho_f(1-\beta(T-T_m))\vec{g}$$

$$\mathrm{div}\,\vec{v} = 0$$

$$\frac{\partial T}{\partial t} = \frac{k_s}{\rho_s c_s}\Delta T$$

$$\left[k\frac{\partial T}{\partial n}\right]^f_s + \rho_f c_f(T-T_m)(\vec{V}_s - \vec{v})\cdot\vec{n} = \rho_s c_s(T-T_m)\vec{V}_s\cdot\vec{n} - \rho_s L V_s$$

$$\frac{T-T_m}{T_m} = -\frac{c_s\rho_s T_m V_s}{L} - \frac{\gamma_s \kappa_s}{L\rho_s} + \left[\frac{k^{-1}}{\rho}\right]^f_s \frac{p_R - p^*}{L}$$

$$\vec{v}(\vec{x}) = \left(1-\frac{\rho_s}{\rho_f}\right)\vec{V}_s, \qquad \vec{x}\in\Gamma_1$$

$$T|_1 = T|_s$$

$$k\frac{\partial T}{\partial n}(\vec{x}) = -h(T(\vec{x})-T^*), \qquad \vec{x}\in\Gamma_3$$

$$-p\vec{n} + \tau\vec{n} = -\gamma_s\kappa_s\vec{n} - p^*\vec{n}$$

$$\vec{v}(\vec{x}) = 0, \qquad \vec{x}\in\Gamma_3$$

$$T(\vec{x}) = T_{F_3}(\vec{x}), \qquad \vec{x}\in\Gamma_3$$

$$k\frac{\partial T}{\partial n}(\vec{x}) = q_{F_3}(\vec{x}), \qquad \vec{x}\in\Gamma_3$$

Correction

Prediction
Overall state

„Overall“ state

Sensor Simulator

$$14)\ \frac{\partial E_t}{\partial(f+g(I_t-I_{t-1}))} = -\chi E_{t-1}^{f+g(I_t-I_{t-1})}\ln E_{t-1} \quad \geq 0,$$

$$15)\ \frac{\partial E_t}{\partial I_t} = -\chi g E_{t-1}^{f+g(I_t-I_{t-1})}\ln E_{t-1} \quad \leq 0,$$

$$16)\ \frac{\partial E_t}{\partial \vec{x}} = -\chi E_{t-1}^{f+g(I_t-I_{t-1})}\ln E_{t-1} \quad \geq 0,$$

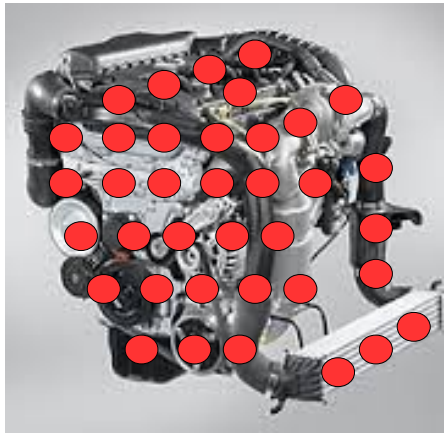
few measurements

comparison&
correction

Weighting:
Model
Sensors

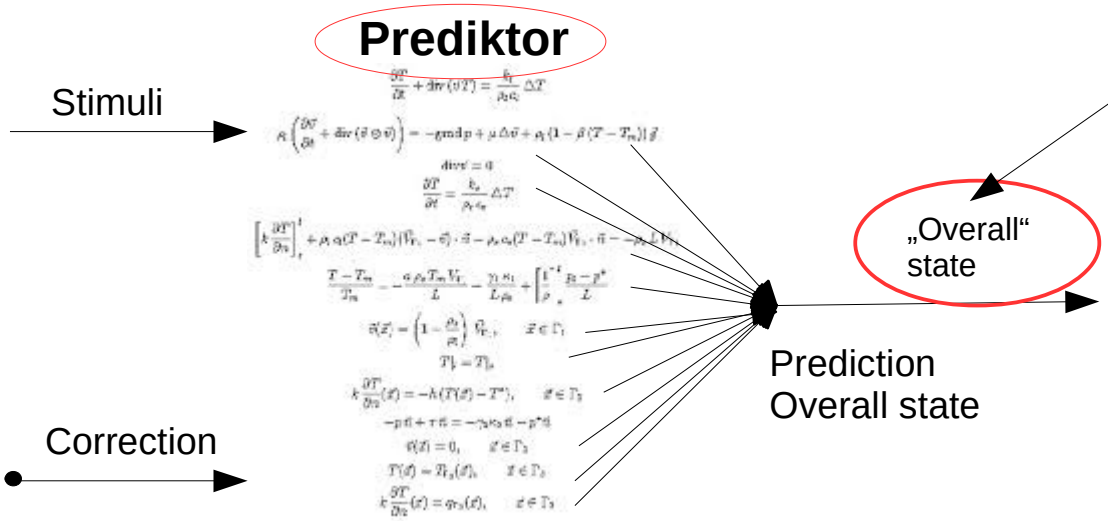
Model + Measurements -> more!

= Temperatures which I „know“



Do I know them?
Where from?

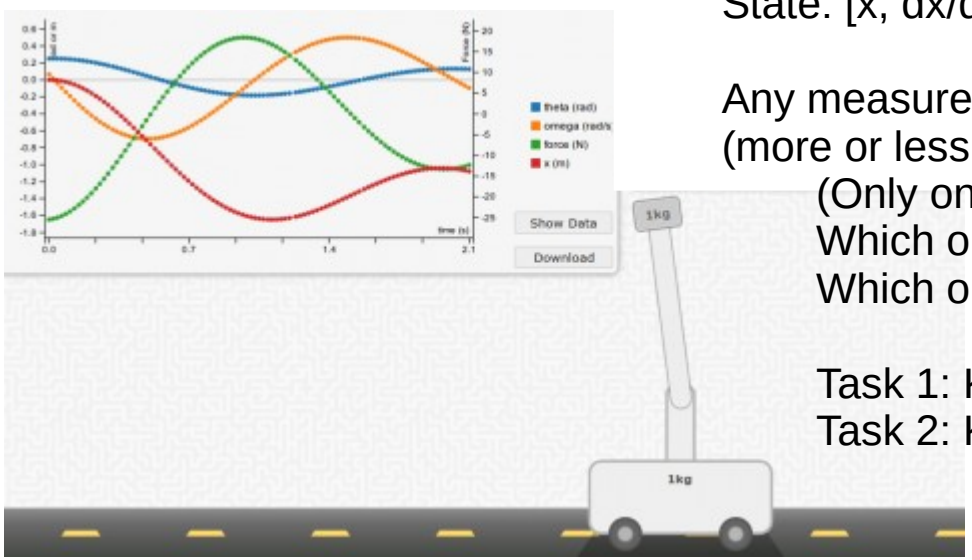
I even „know“ the pressure inside the cylinder all the time... Without sensor!



The more I know about the whole state,
the better I can control.

More Visibility with few Sensors

I measure position, so I can calculate V & A
With this model, I only need one sensor and get complete „Observability“



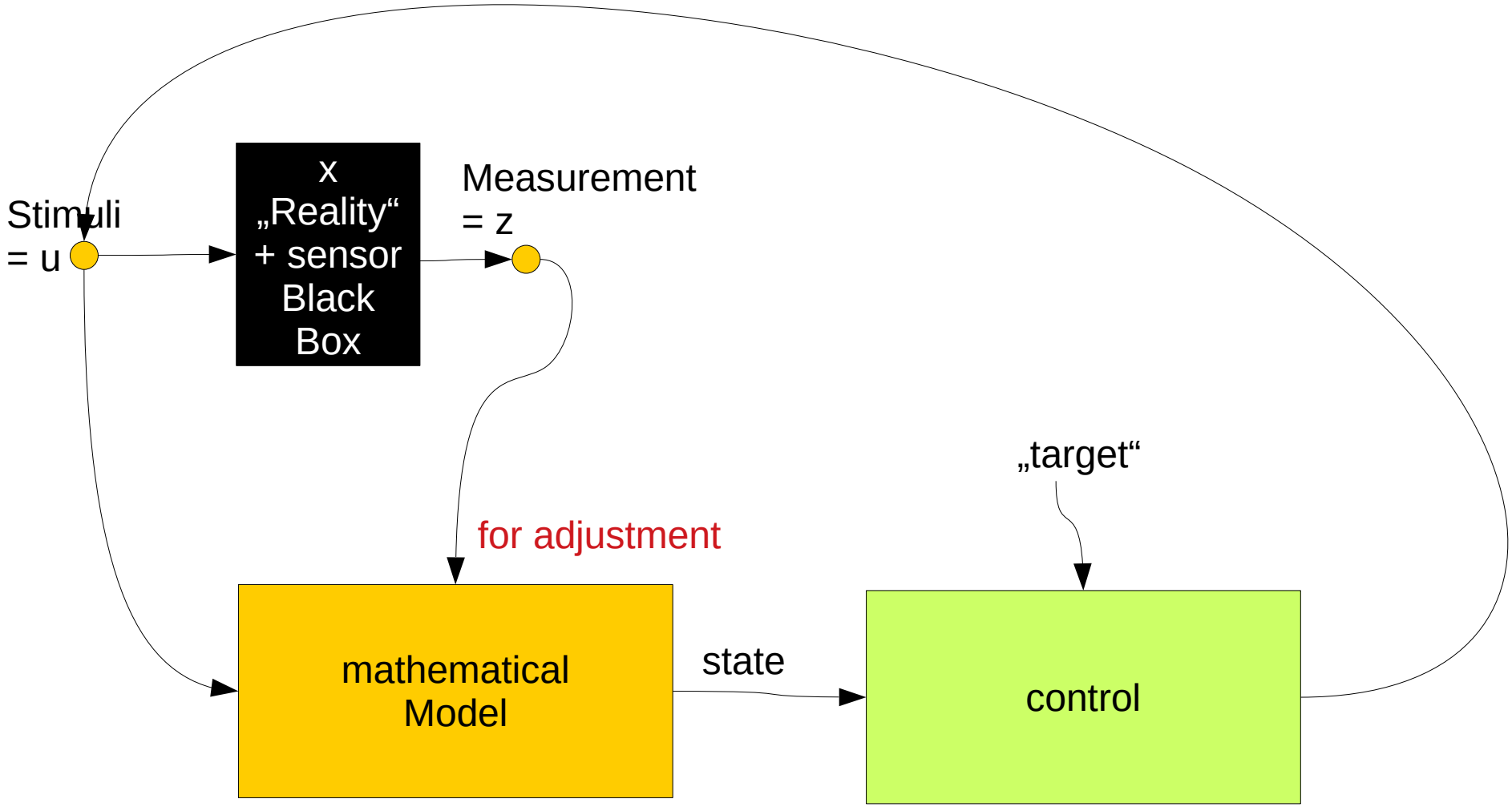
Inverse pendulum: The classic example of control

State: [x, dx/dt, α, dα/dt] … Do I need morer? Then calculate

Any measurement is enough to know the state completely
(more or less… which measurement would help to determine the rest?
(Only one sensor!)
Which one is simpler?
Which one is more difficult/expensive/sensitive?

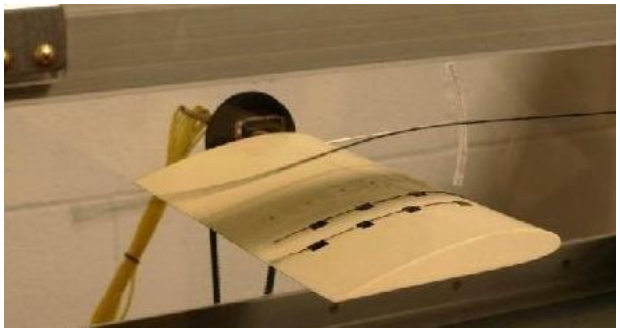
Task 1: Keep pendulum vertically
Task 2: Keep pendulum vertically and cart near position 0

Model
(a Simulation)

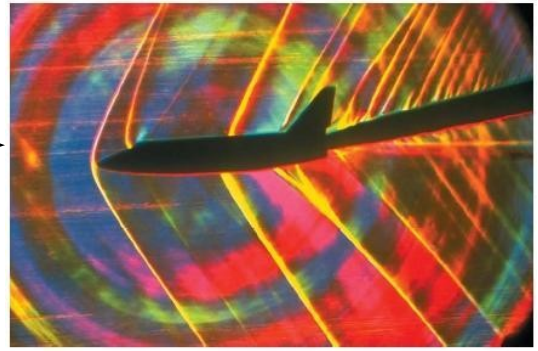


Model („knowing everything“)

The model is derived from design dokuments + engineering experience +
1000x measuremants and experiments,
from much more sensors, that I have in flight: internal + external + very big and expensive



Result:
The modell, that can tell me
„what if“
before I do it

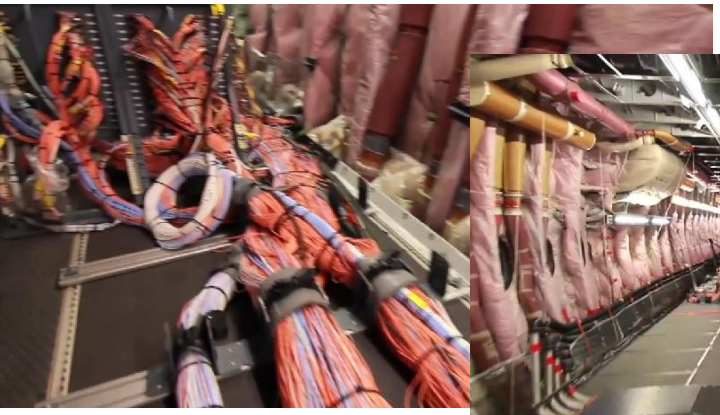


Model Verification

20 metric tons of additional equipment!

A380

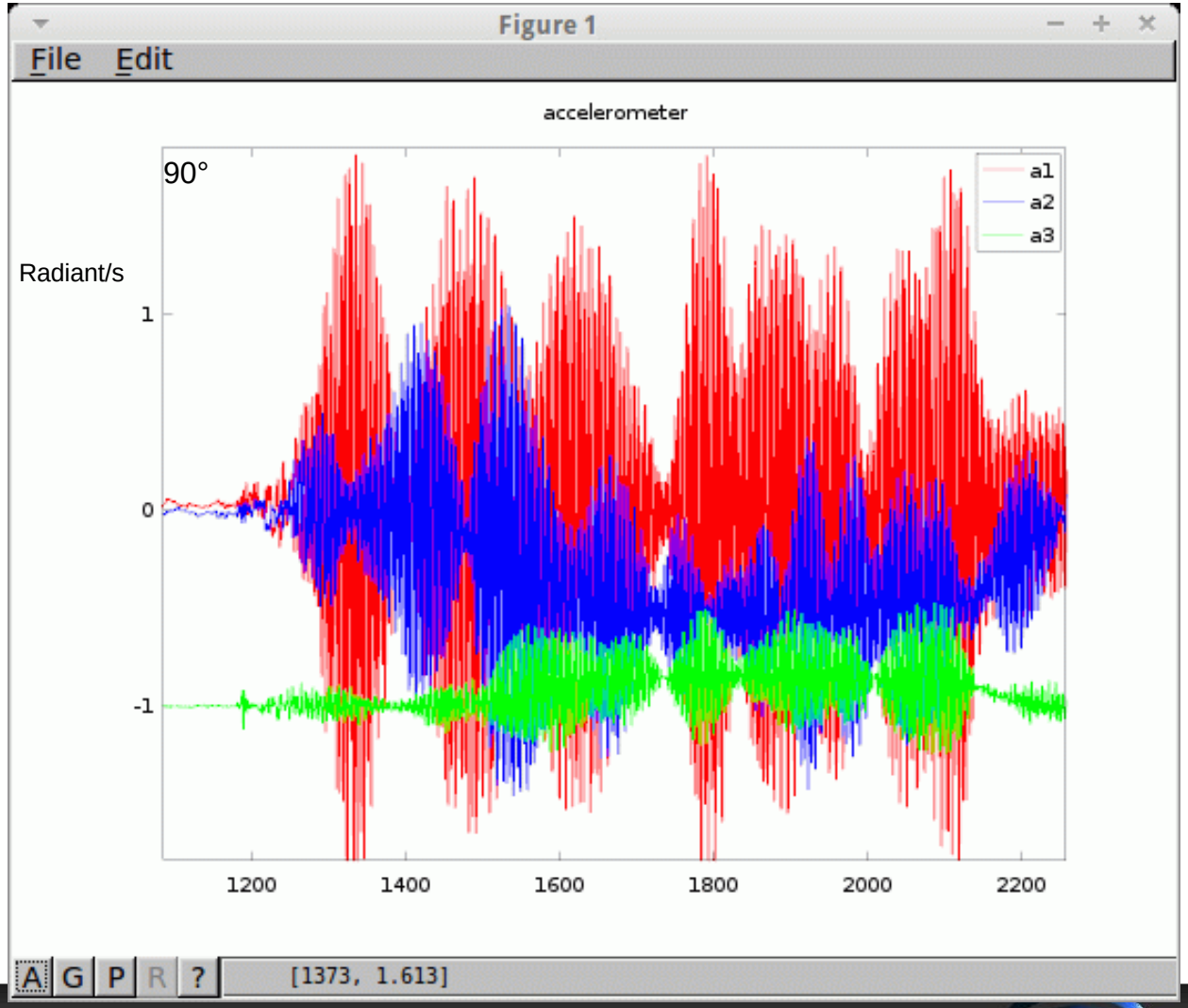
A 350



Flow measurement
Wing A340

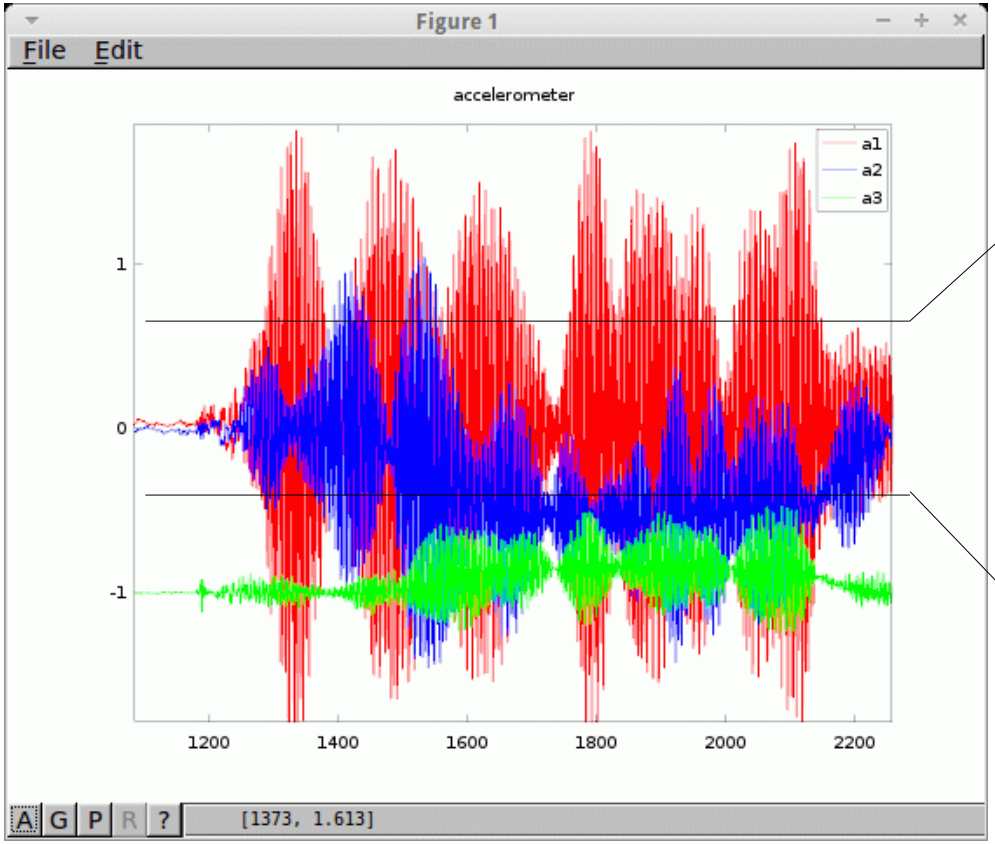
Measurement only: Unusable

Gyros: 3D Rotation rate

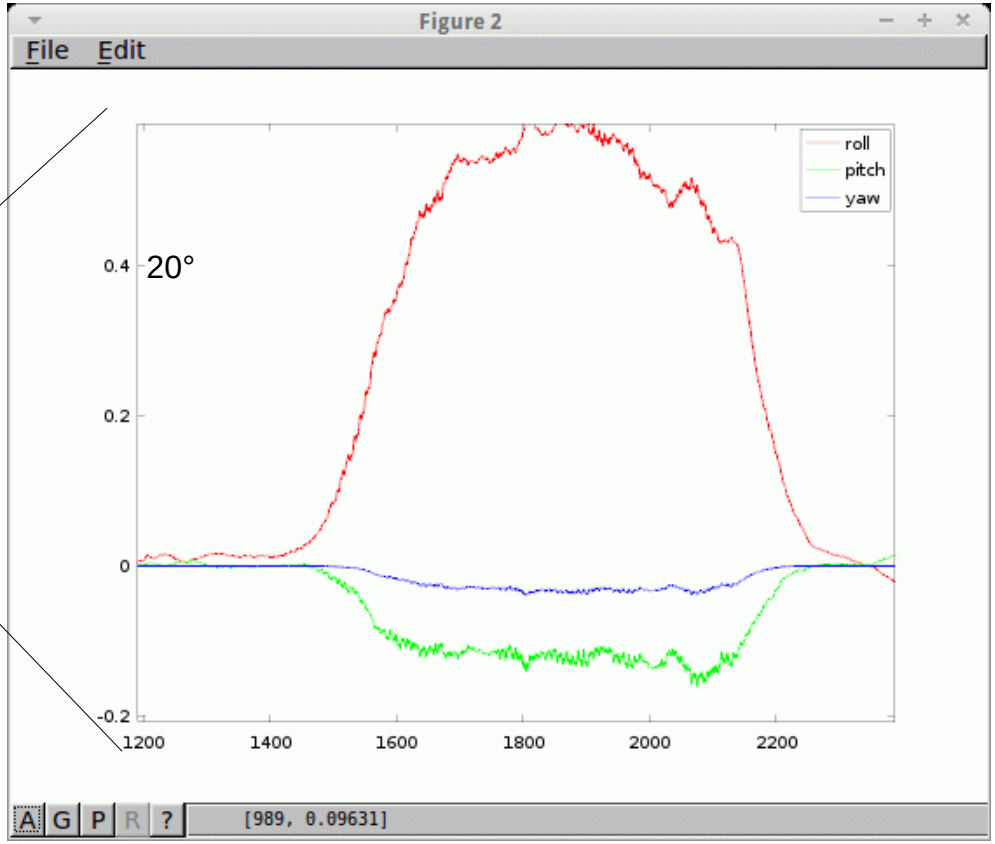


Do I Find Information in these Data (Measurements)?

Gyro in our quadrocopter



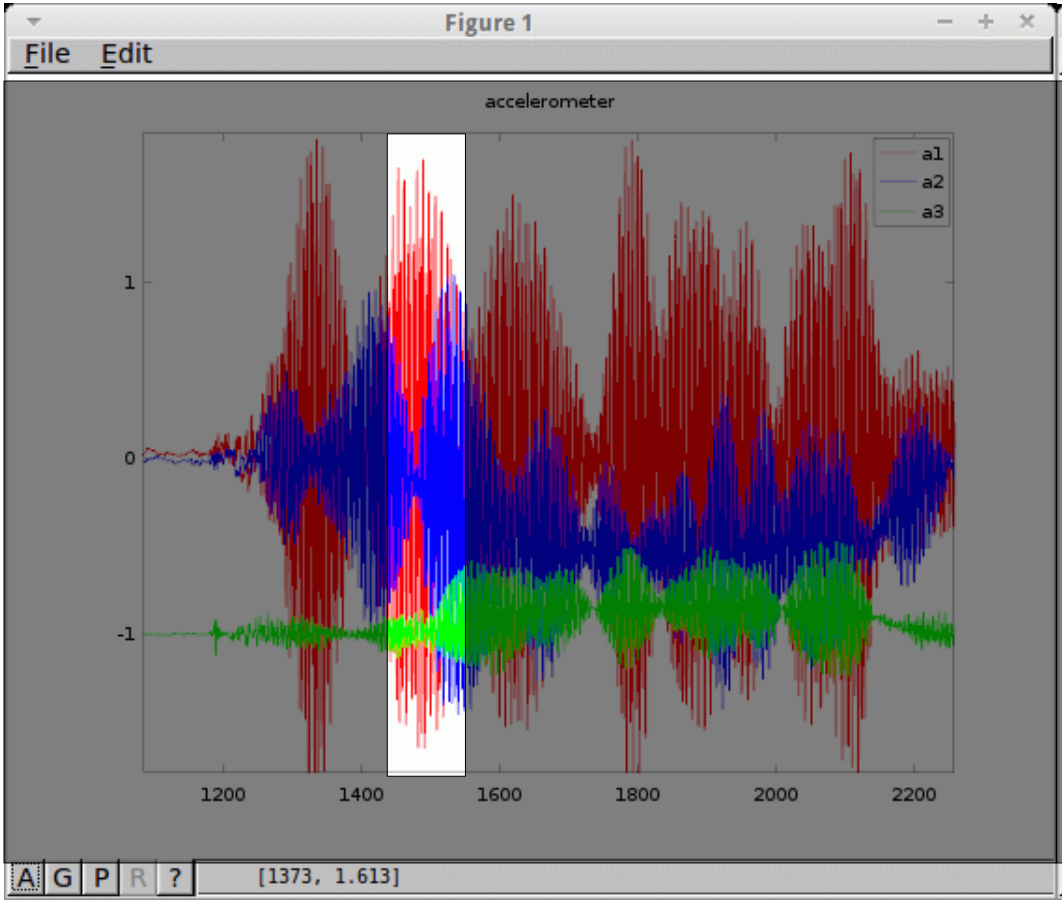
Low pass filter + Kalman (for computer scientists)



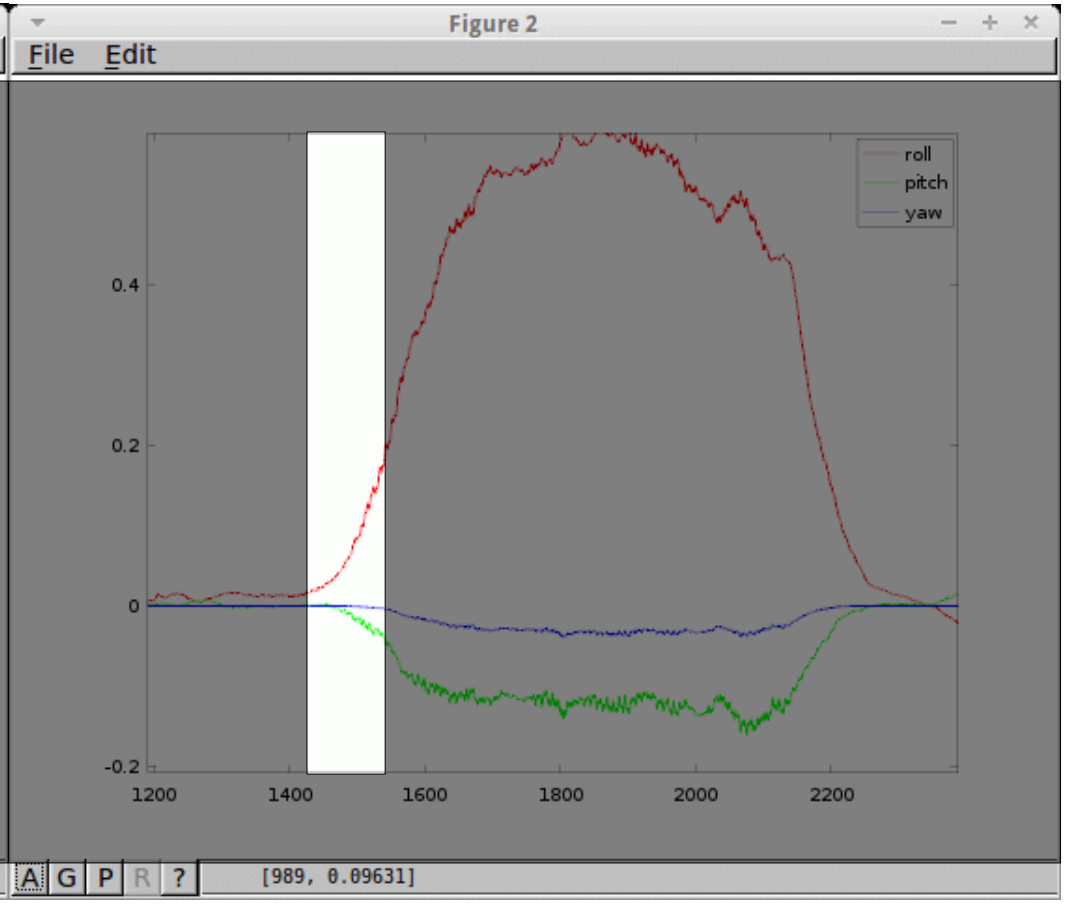
= „Performance“ of filter

Do I Find Information in these Data (Measurements)?

Gyro in our quadrocopter

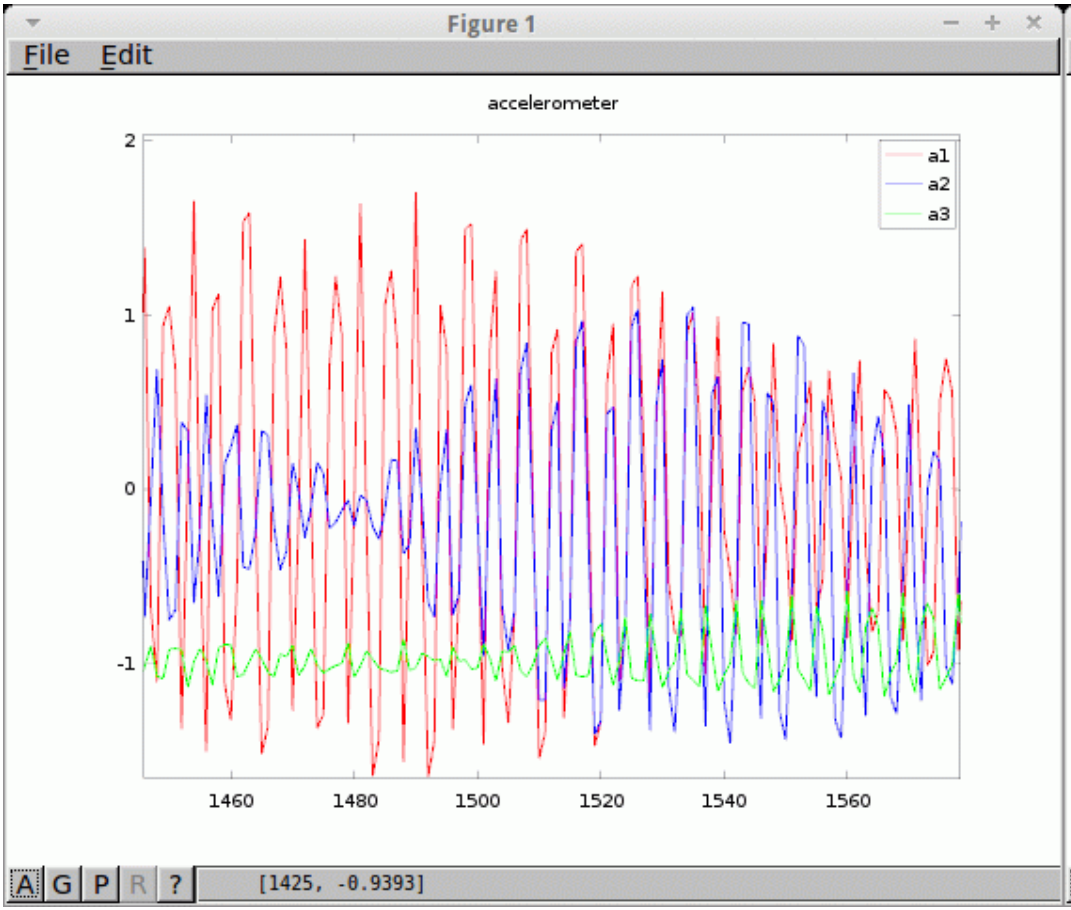


Low pass filter + Kalman (for computer scientists)

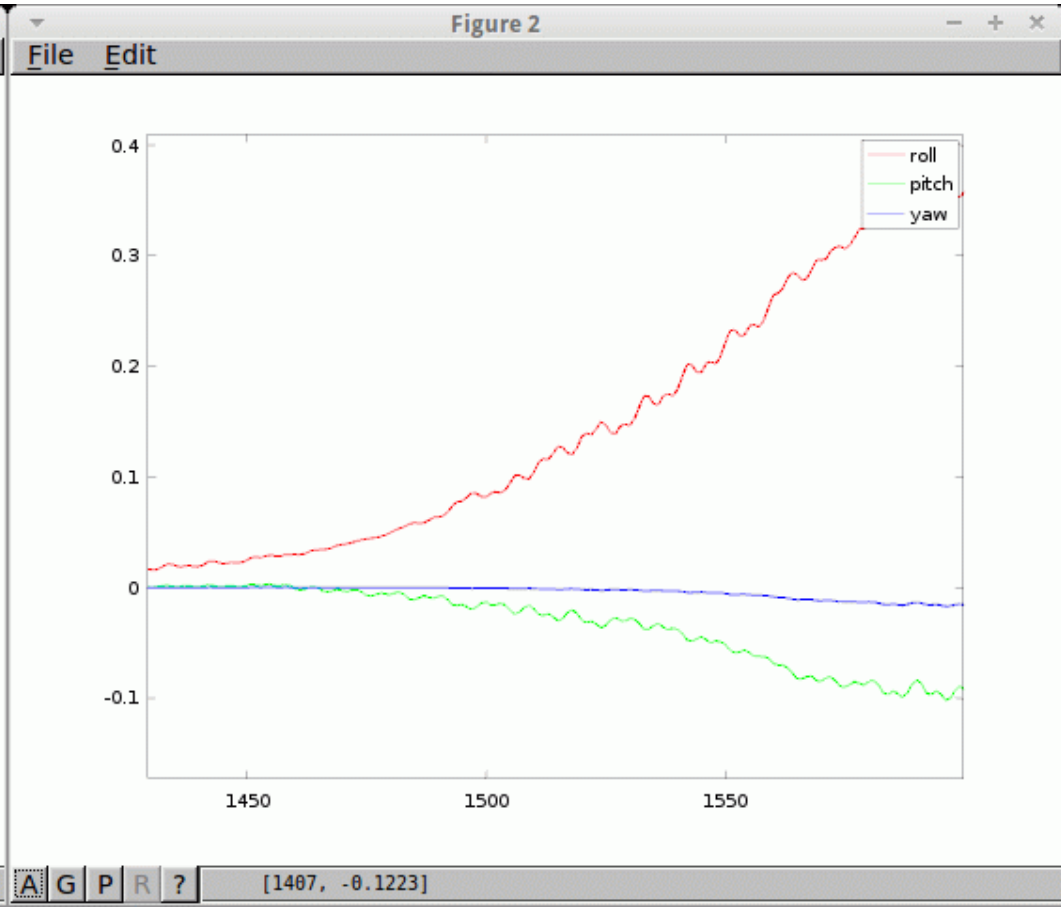


Do I Find Information in these Data (Measurements)?

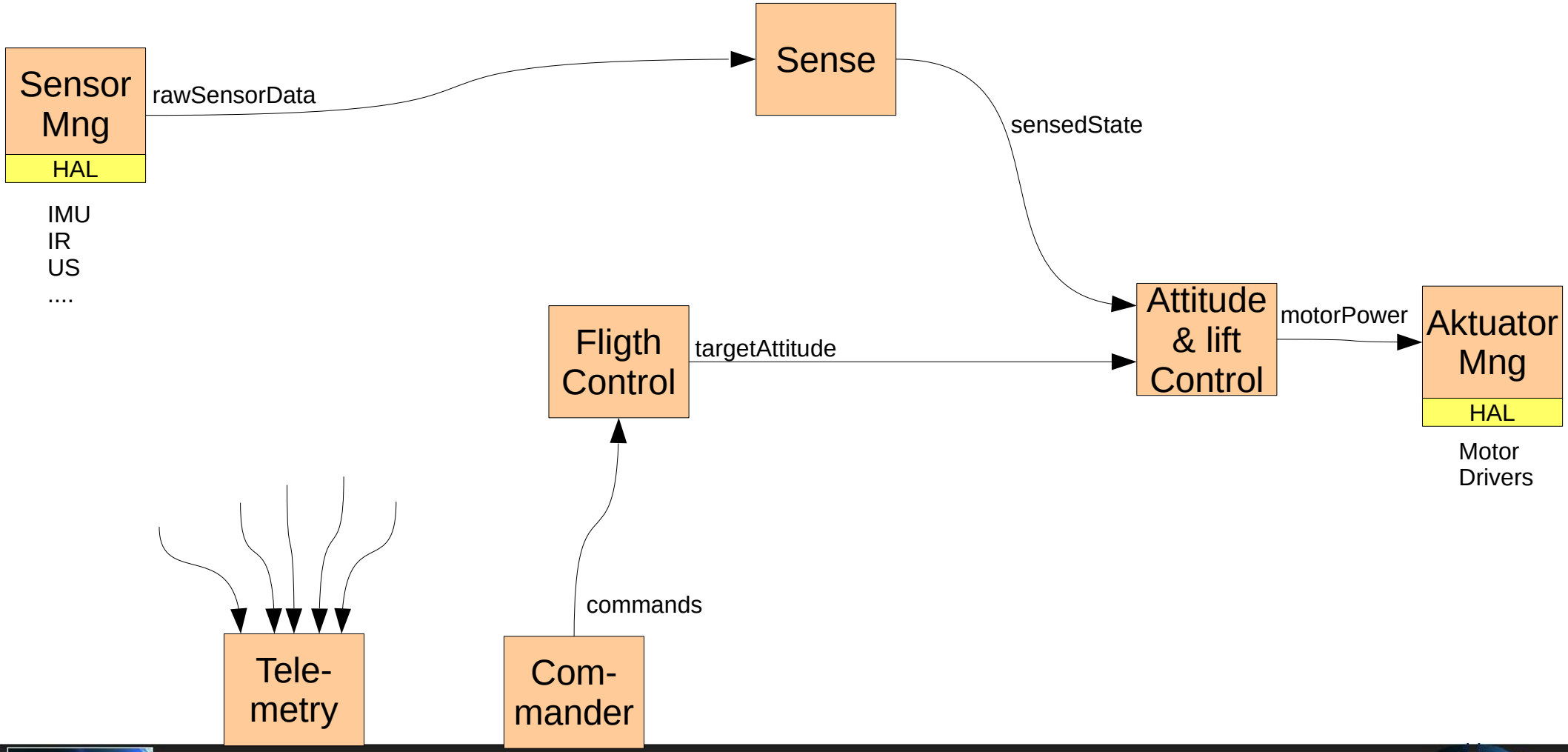
Gyro in our quadrocopter



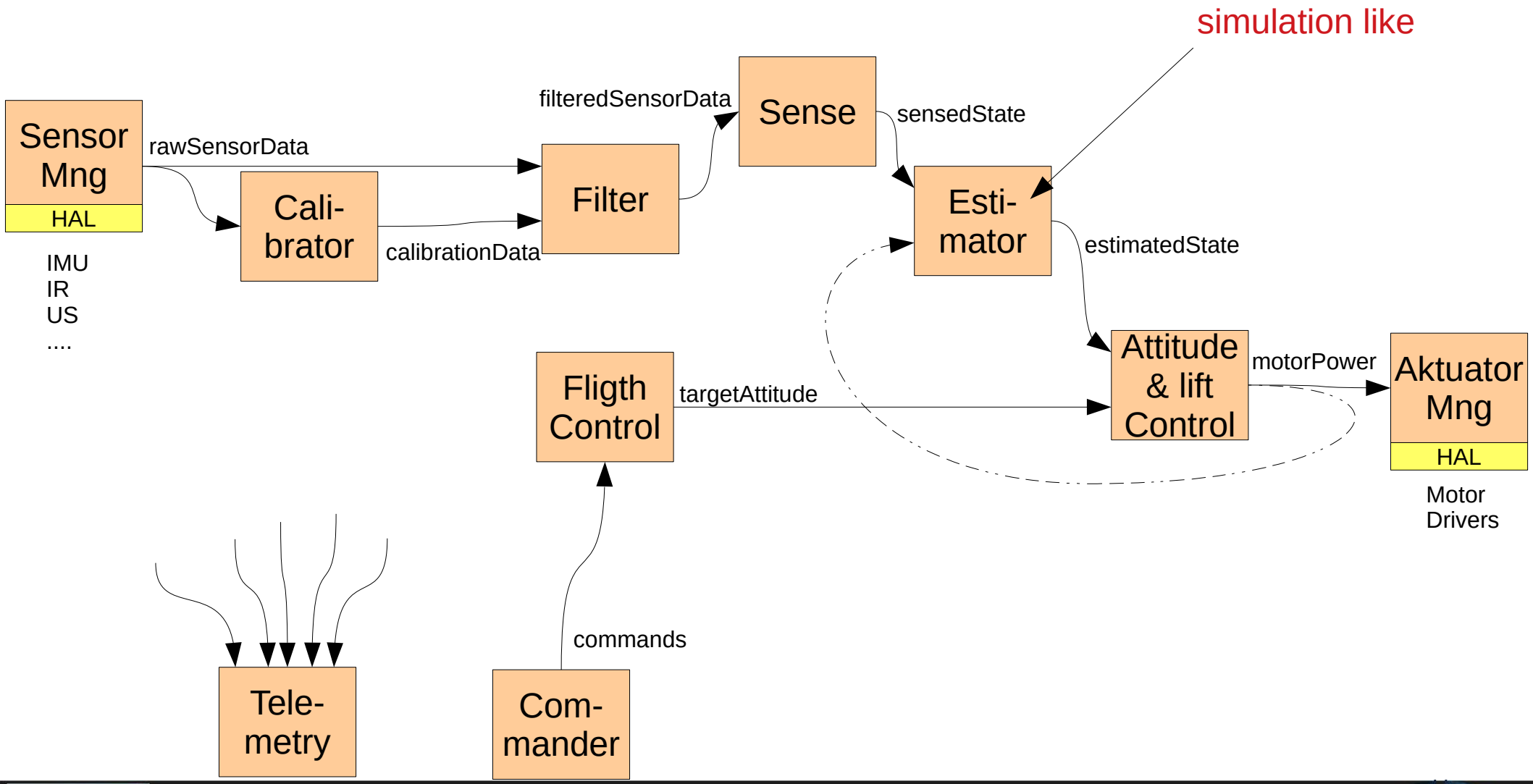
Low pass filter + Kalman (for computer scientists)



Most Simple Control



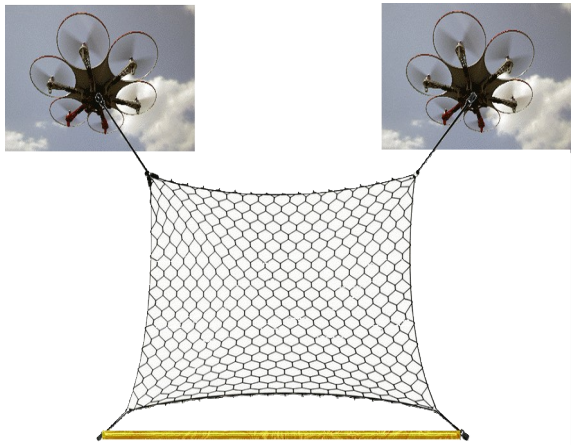
Typical „good“ Control



Viewing the Future…

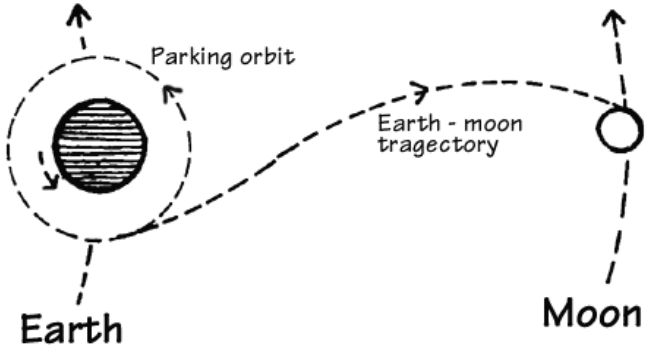
Sometimes I have to view the future... What will be

If I know how the system is going to behave, I can find out what I have to adjust to make it do what I want.



MIDRAS: Interceptor drones

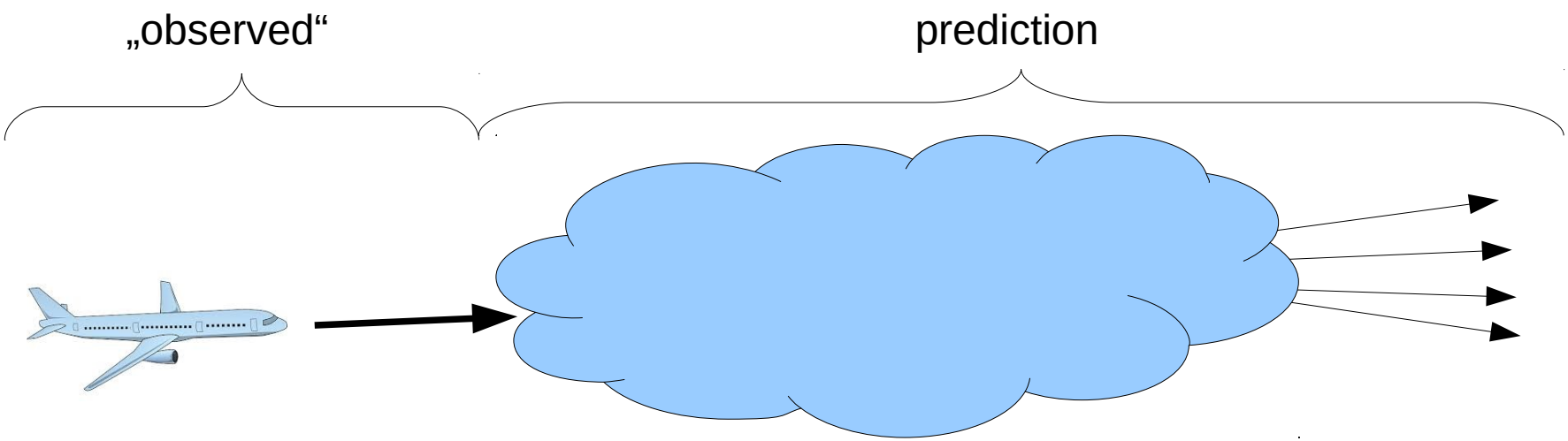
Fly to the moon?
-> not where it is now, but where it will be when I arrive
-> where shall I go



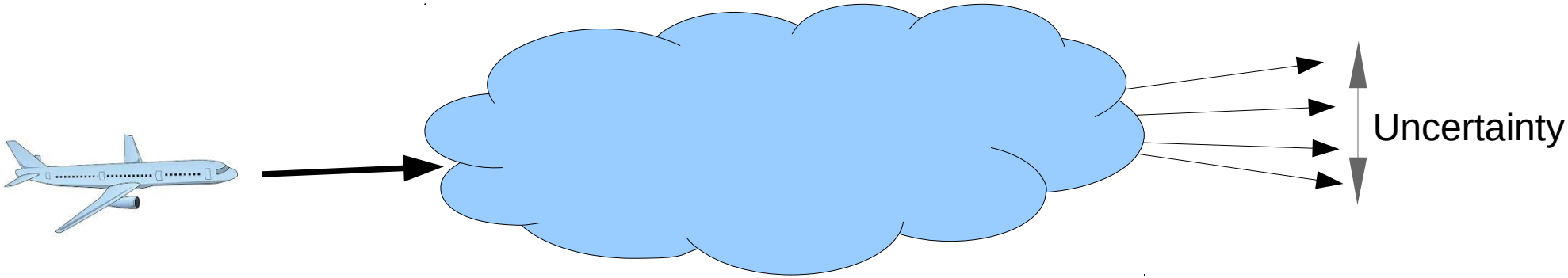
„Trafic“ alert

ads-b

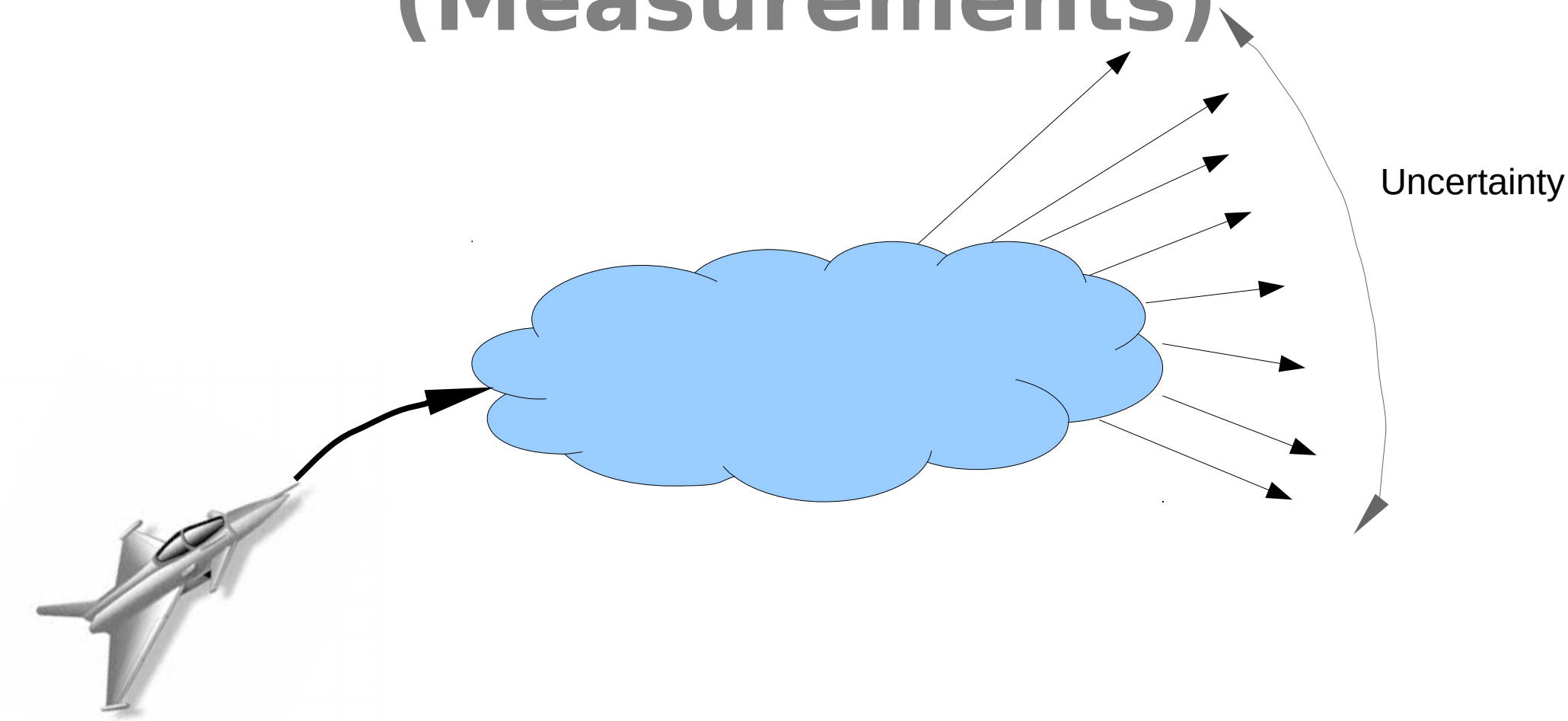
Model and Observations (Measurements)



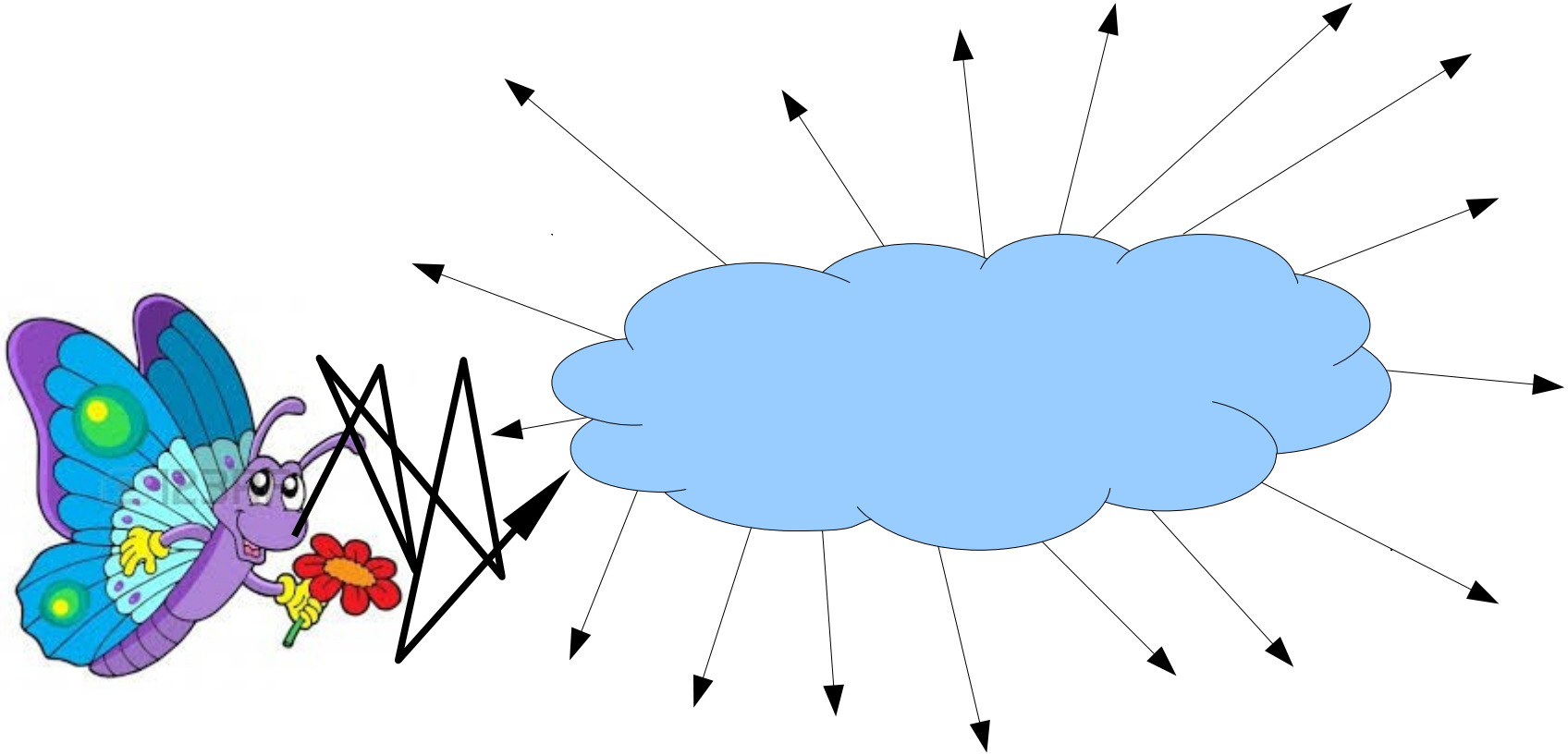
Model and Observations (Measurements)



Model and Observations (Measurements)



Model and Observations (Measurements)

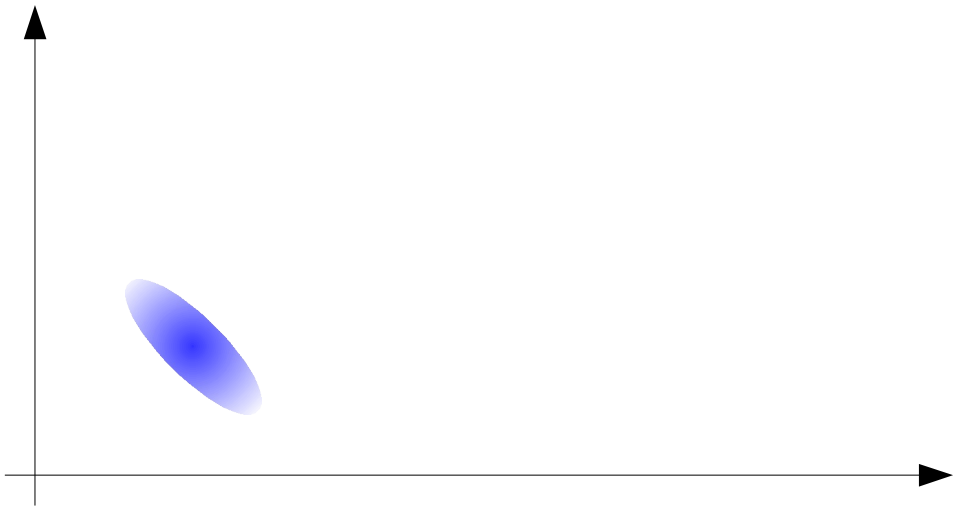


Certainty??

Our Task

Calculate with the unpredictable
Calculate with uncertainty

Measure and estimate

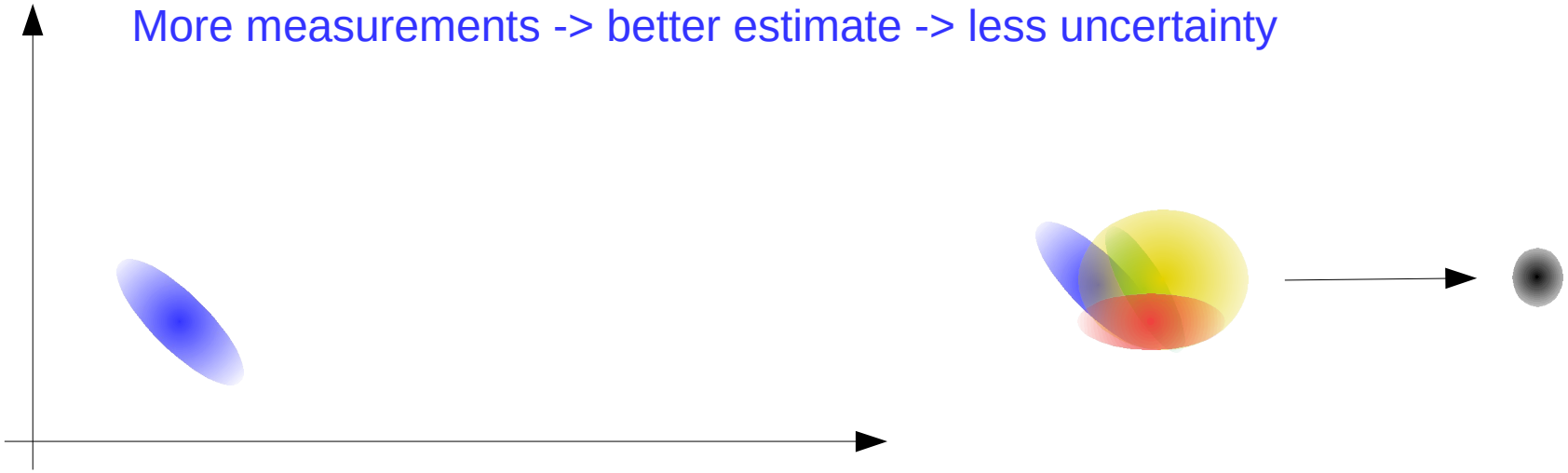


Cloud of probability

Sensor Fusion

People as sensor: Temperature in this room?
more measurements, more exact? nX same person, or 1 x n X persons?

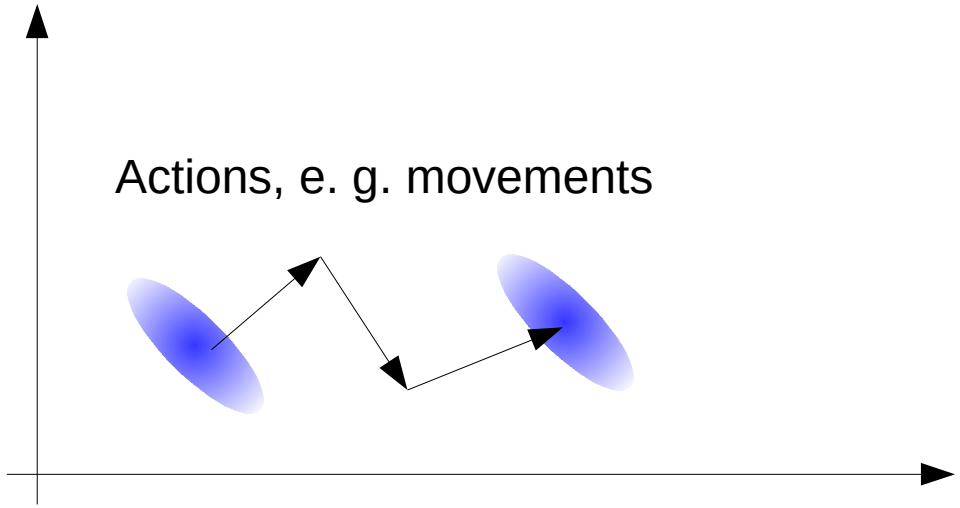
More measurements -> better estimate -> less uncertainty



Cloud of probability

Actions

Estimate (people as sensor) my position:



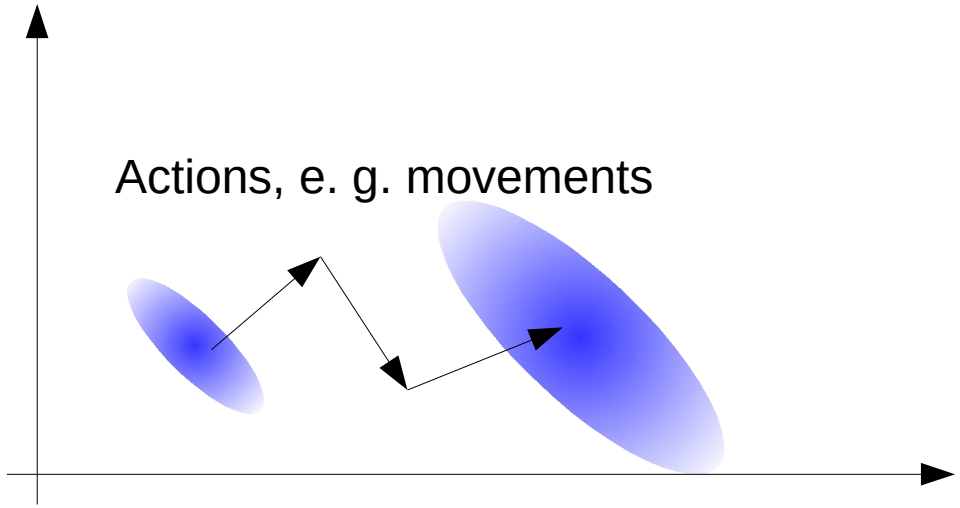
final position =
start position + all movements

e. g. 90cm right, 50cm straight ....

Same Uncertainty as my start position?

Messen und Aktionen

Estimate (people as sensor) my position:



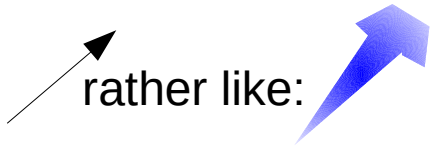
final position =
start position + all movements
e. g. 90cm right, 50cm straight ....

Same Uncertainty as my start position?

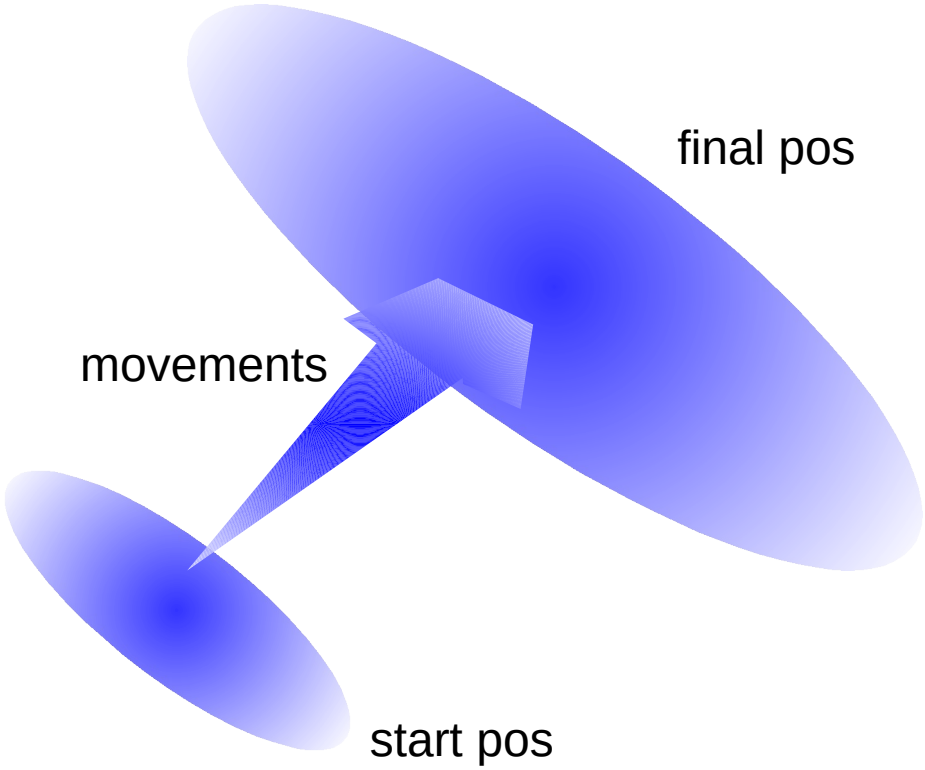
Actions worsen my estimate -> greater uncertainty

every movement has uncertainty

rather like:



Actions



I add up all the movements
then also all the uncertainties

Fusion und Concatenation

Fusion for best estimation
(Estimator)

= Multiplication of distributions
Measurement a with distribution A
Measurement b with distribution B

Kombination:
C = A*B

Concatenation of states + actions
z.B. Position + movement
or attitude and roation rate

= Sum of distributions
state of start a with distribution A
Action b with distribution B

Concatenation:
C = A+B

if A and B are normal distribution -> C too

Normal distribution = N(μ,v)
but I prefer
weighted values = (μ, G) = (μ, 1/v)

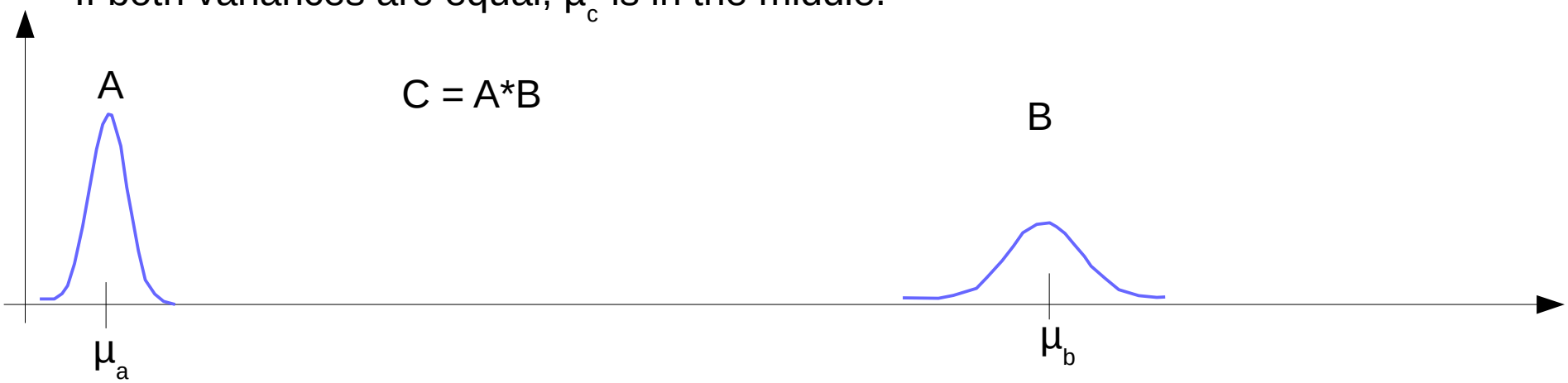
Multiplication of Normal Distributions in the same Dimension

$$Contour = f(x, y) = \left(\frac{e^{-(x-\bar{x})^2/2*v^2}}{\sqrt{2*\pi*v}}\right)*\left(\frac{e^{-(y-\bar{y})^2/2*v^2}}{\sqrt{2*\pi*v}}\right)$$

Here: x and y are two values in the same dimension

C = A*B;

μ_c = weighted average of μ_a and μ_b
μ_c is closer to the multiplier with the smallest variance (weight = 1/v = certainty).
If both variances are equal, μ_c is in the middle.



Multiplication of Normal Distributions in the same Dimension

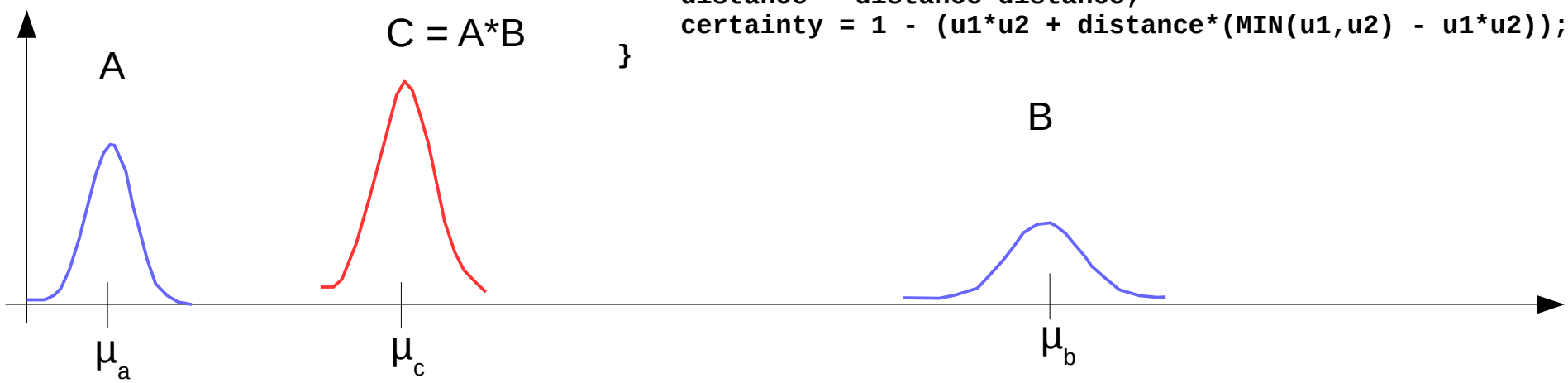
C = A*B;
Sufficient approximation: weighted values = (μ, G) = (μ, 1/v)

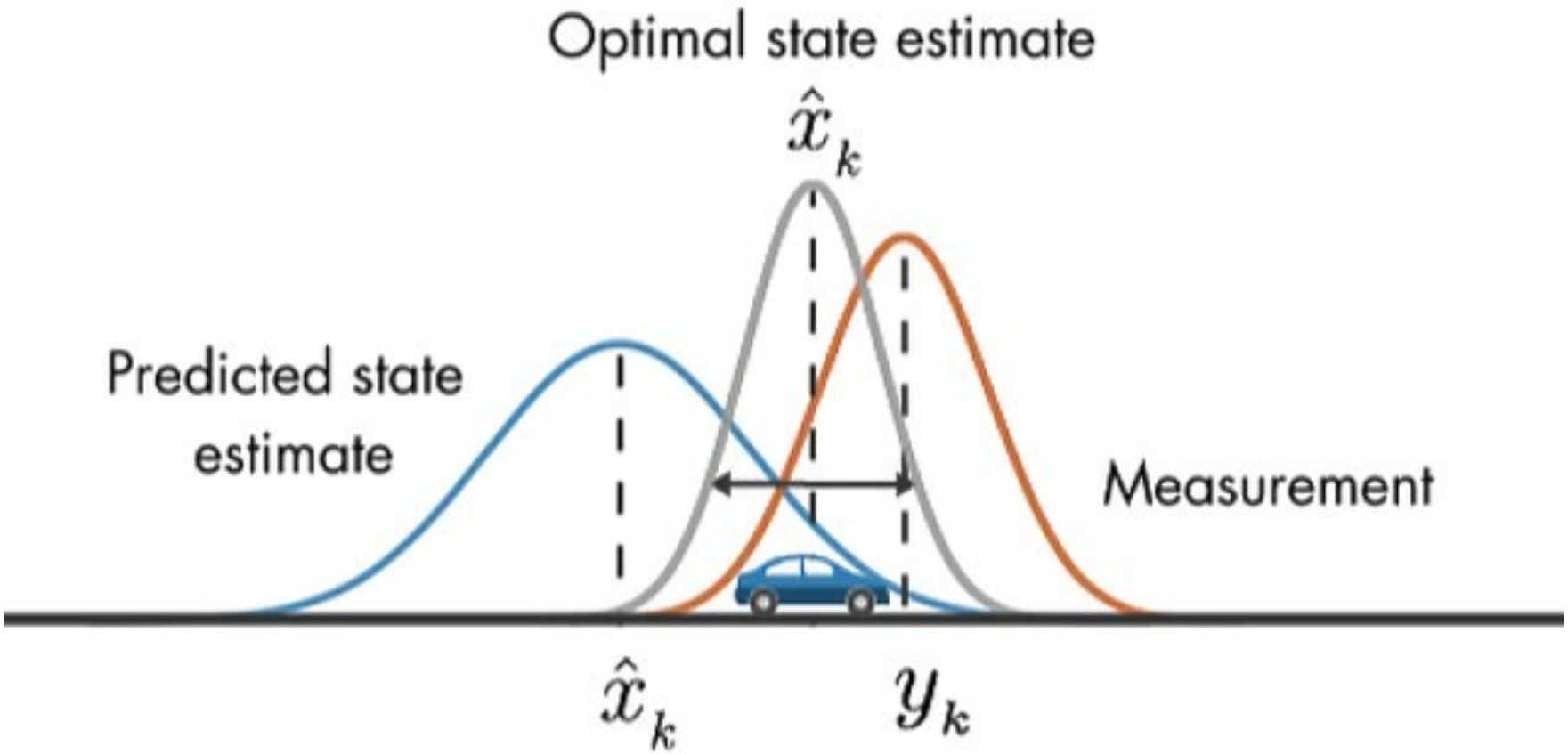
val certainty1

$$\mu_c = \frac{\mu_a * G_a + \mu_b * G_a}{G_a + G_b}$$

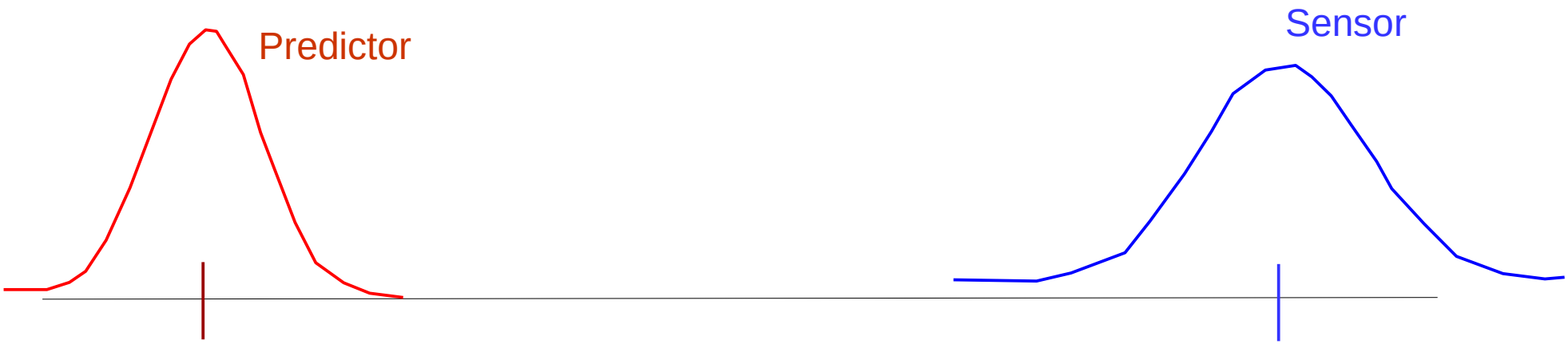
```
inline void ThrustedVal::MultiplyValAndCertainty() {
    mergeVal();

    // increase certainity according to conformity of both values
    // easier to compute using uncertainty u = (1 - certainty).
    // range of new uncertainty = u1*u2 ... min(u1,u2)
    // smallest uncertainty (u1*u2) if both values are identical
    // biggest uncertainty (mnin(u1,u2) if values diverge by a lot
    double u1 = 1 - certainty1;
    double u2 = 1 - certainty2;
    double distance = (val1 - val2);
    distance = distance*distance;
    certainty = 1 - (u1*u2 + distance*(MIN(u1,u2) - u1*u2));
}
```

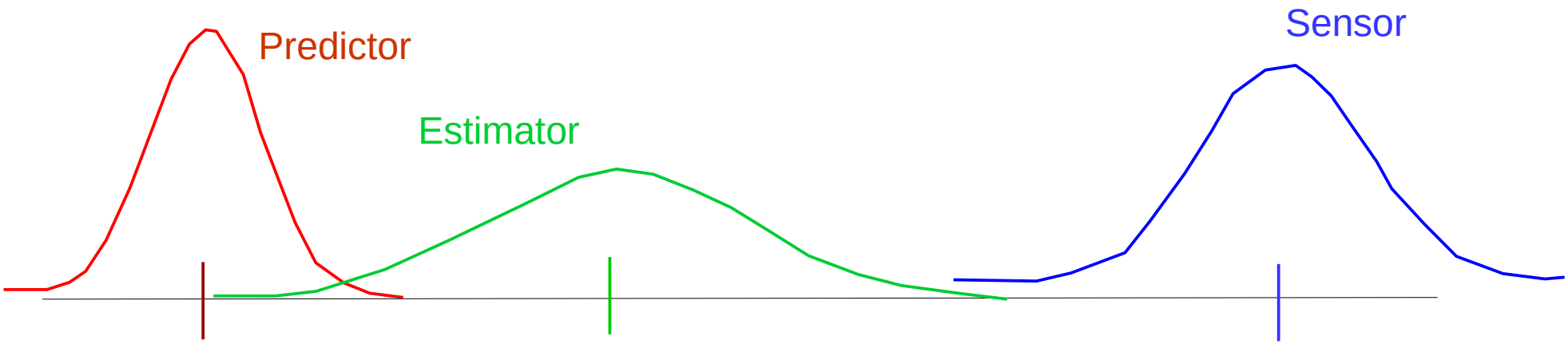



Optimal state estimate
$\hat{x}_k$
Predicted state estimate
Measurement
$\hat{x}_k$
y_k

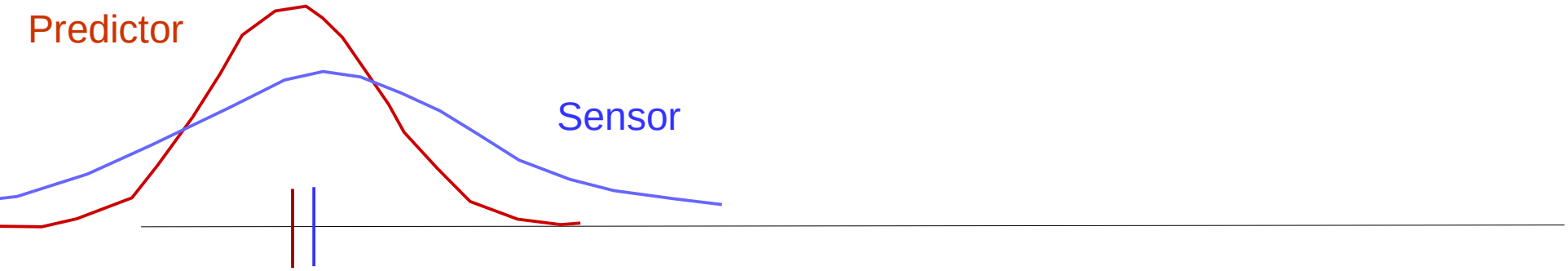
Fusion
Estimator = Predictor * Sensor



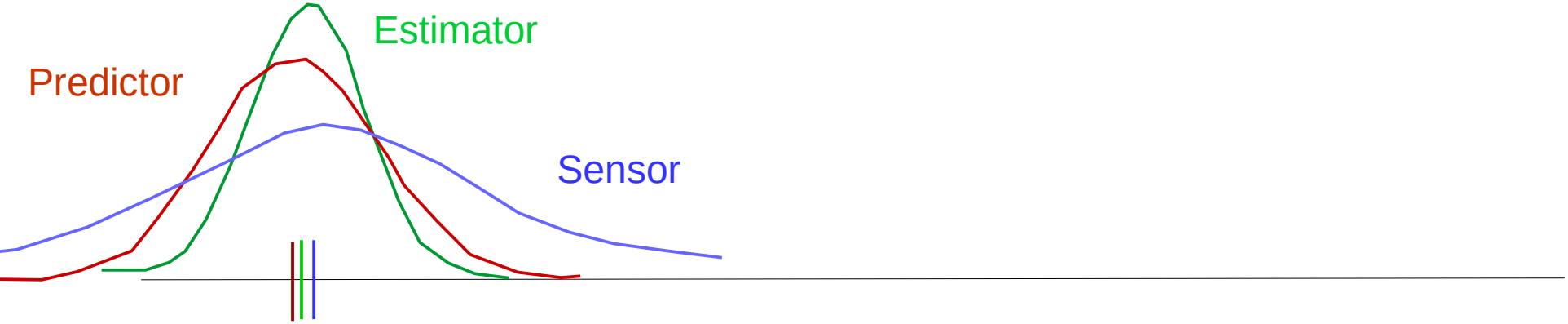
Fusion
Estimator = Predictor * Sensor



Fusion Sensor + Model

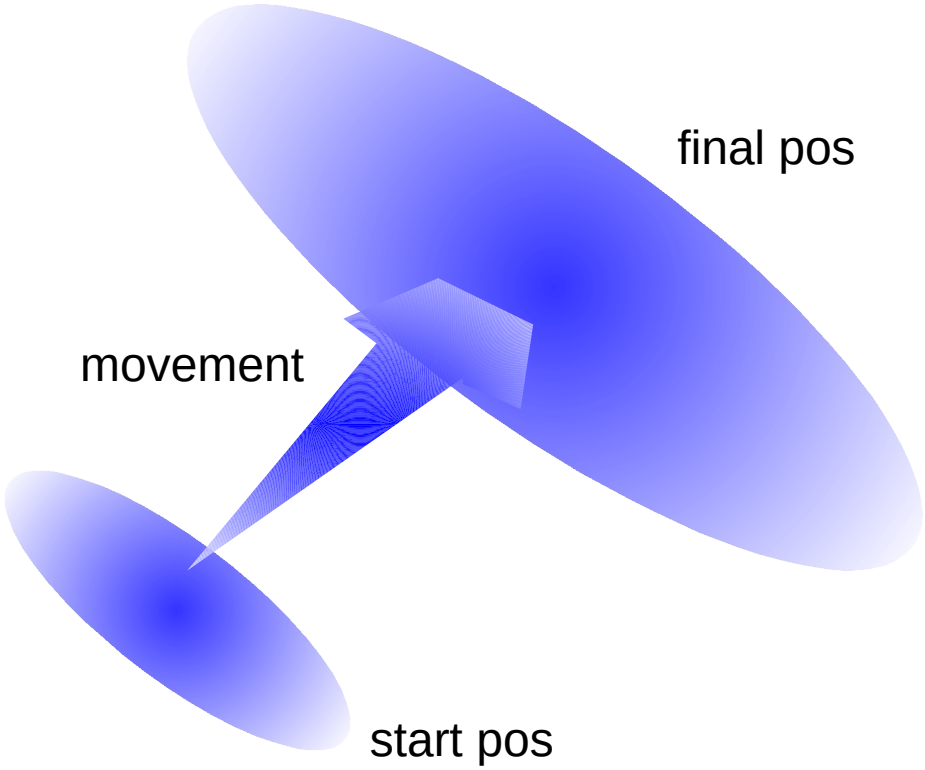


Fusion
Estimator = Predictor * Sensor



see 035_merge_normal_distribution-slow.m

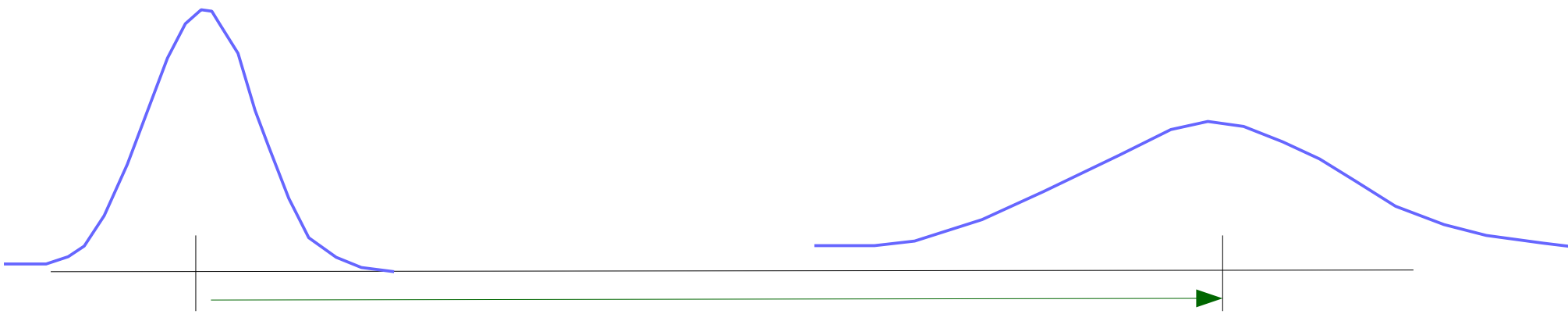
Actions : Concatenation
C = A + B



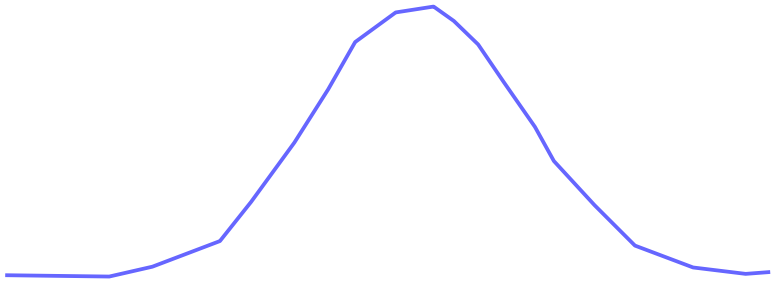
Actions : Concatenation
C = A + B

x(t-1) : Position with uncertainty

x(t) : Position with uncertainty



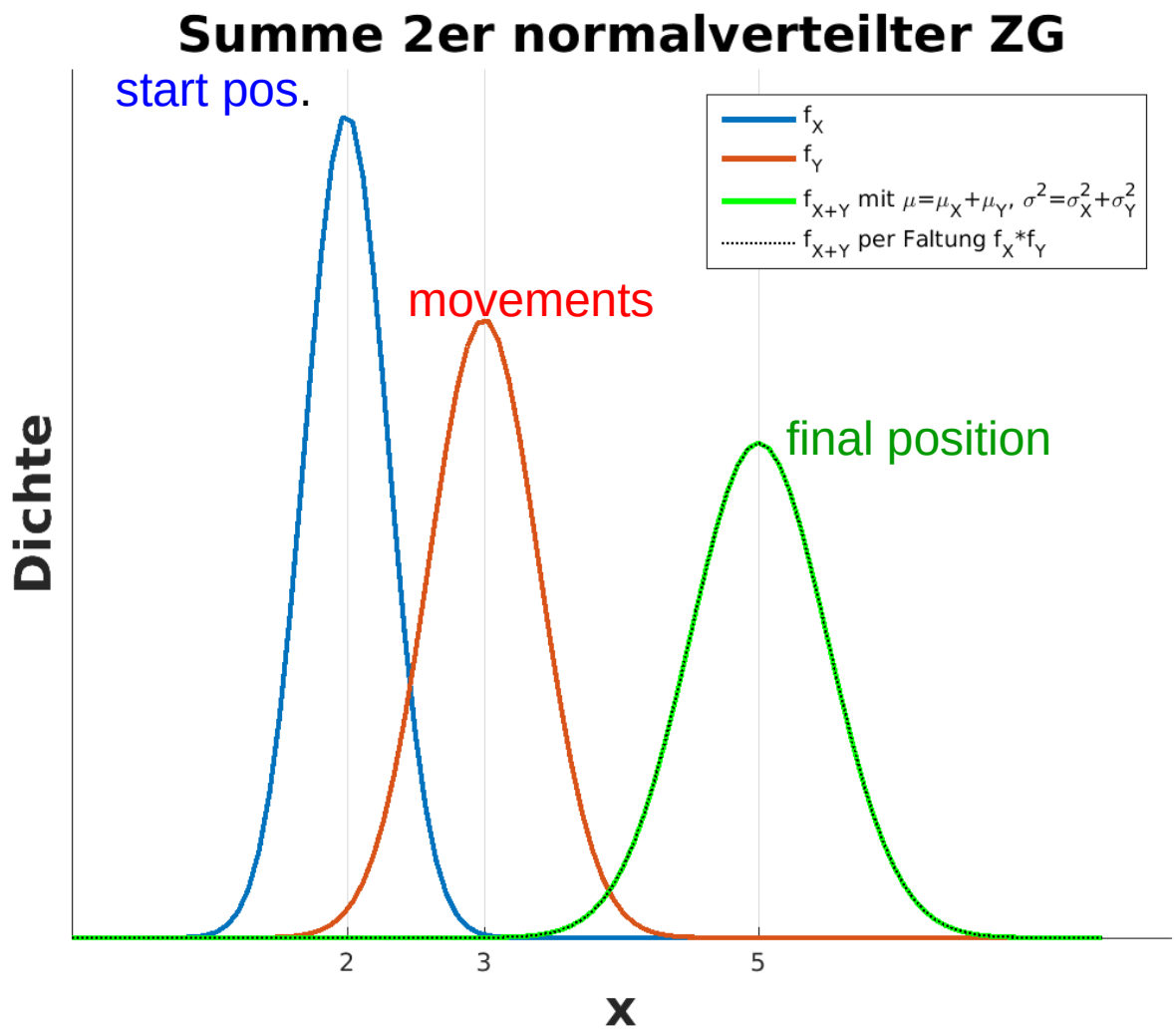
(estimated speed) * (estimated time)



The more I continue
the more my uncertainty

Move or rotate with your
eyes closed

Concatenation Sum



$$\mu_c = \mu_a + \mu_b$$

$$v_c = v_a + v_b$$

Remember:
Neither model nor measurements are perfect
(far from it)

We are able to improve measurements with filter,

but best
-> we use them both: model & measurement
    -> e. g. Kalman filter
        (is actually not a filter, but an estimator)

And... then we get everything only roughly!
    (best knowledge and belief)
        Humans also do everything roughly and it
        works just fine!

NxSensor Fusion + Model Fusion → greater certainty
Concatenation of actions → less certainty